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# Notes

A. van Hameren

*Institute of Nuclear Physics Polisch Academy of Sciences*
*PL-31342 Kraków, Poland*

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**Abstract**

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## 1 Notation

We decompose a general four vector $K^\mu$ as

$$K^\mu = x_K P^\mu + \bar{x}_K \bar{P}^\mu + K_\perp^\mu \tag{1}$$

where

$$P^2 = \bar{P}^2 = 0 \quad , \quad 2P\cdot\bar{P} = s > 0 \quad , \quad P\cdot K_\perp = \bar{P}\cdot K_\perp = 0 \tag{2}$$

and

$$x_K = \frac{\bar{P}\cdot K}{P\cdot\bar{P}} \quad , \quad \bar{x}_K = \frac{P\cdot K}{P\cdot\bar{P}} \, . \tag{3}$$

The coordinates $x$ and $\bar{x}$ are dimensionless and strictly positive for positive-energy momenta. In terms of these coordinates, the infinitesimal volume of a light-like momentum becomes

$$\begin{aligned} d^4K\,\delta(K^2) &= (P\cdot\bar{P})\,dx_K d\bar{x}_K d^2K_\perp\,\delta\left(2(P\cdot\bar{P})x_K\bar{x}_K - |K_\perp|^2\right) \\ &= dx_K d\bar{x}_K d^2K_\perp\,\frac{1}{2x_K}\,\delta\left(\bar{x}_K - \frac{|K_\perp|^2}{sx_K}\right) \, . \end{aligned} \tag{4}$$

## 2 High energy limit

Consider the integral

$$I \;=\; \int_0^\infty dx_1 \int_0^\infty dx_2\, \delta(x_1+x_2-\Lambda)\, \frac{1}{\Lambda}\, \frac{x_1}{(a+x_2)^2} \;=\; \frac{1}{a} + \frac{1}{\Lambda} \ln \frac{a+\Lambda}{a} \;=\; \frac{1}{a} + \mathcal{O}\big(\Lambda^{-1}\big)\,, \tag{5}$$

where both $a > 0$ and $\Lambda > 0$, but we consider $\Lambda$ to be large. The constraint $\delta(x_1 + x_2 - \Lambda)$ demands $\Lambda$ to be distributed over $x_1$ and $x_2$. The integrand suggest that $x_1$ should "get most of it", because it vanishes if $x_2 \to \Lambda \to \infty$. Let us try to find the result of the integral up to order $1/\Lambda$ by making use of the observations before. We use that $1 = \int_{-\infty}^{\infty} dy\, \delta(y + x_1 - \Lambda)$ and write

$$\begin{aligned} I &= \int_{-\infty}^{\infty} dy \int_0^\infty dx_1 \int_0^\infty dx_2\, \delta(y+x_1-\Lambda)\, \delta(x_1+x_2-\Lambda)\, \frac{1}{\Lambda}\, \frac{x_1}{(a+x_2)^2} \\ &= \int_{-\infty}^{\Lambda} dy \int_0^\infty dx_2\, \delta(y+x_2)\, \frac{1}{\Lambda}\, \frac{\Lambda - y}{(a+x_2)^2}\,. \end{aligned} \tag{6}$$

Now, we apply the limit $\Lambda \to \infty$ on the integrand, to find

$$I \;\to\; \int_{-\infty}^{\infty} dy \int_0^\infty dx_2\, \delta(y+x_2)\, \frac{1}{(a+x_2)^2} \;=\; \frac{1}{a}\,. \tag{7}$$

The essence of this exercise is that we would have found the wrong result had we introduced $1 = \int_0^\infty dy\, \delta(y + x_2 - \Lambda)$ instead of the one with $x_1$, and then taken the limit on the integrand.

More general, let $f_\Lambda(x_1, x_2, \ldots, x_n)$ be finite if $x_1 \to \Lambda \to \infty$, but vanish for this limit on any other argument. Then we can write

$$\begin{aligned} I &= \int_0^\infty dx_1 dx_2 \cdots dx_n\, \delta(x_1+x_2+\cdots+x_n-\Lambda)\, f_\Lambda(x_1,x_2,\ldots,x_n) \\ &= \int_{-\infty}^{\infty} dy \int_0^\infty dx_1 dx_2 \cdots dx_n\, \delta(y+x_1-\Lambda)\, \delta(x_1+x_2+\cdots+x_n-\Lambda)\, f_\Lambda(x_1,x_2,\ldots,x_n) \\ &= \int_{-\infty}^{\Lambda} dy \int_0^\infty dx_2 \cdots dx_n\, \delta(y+x_2+\cdots+x_n)\, f_\Lambda(\Lambda-y,x_2,\ldots,x_n) \\ &= \int_{-\infty}^{\infty} dy \int_0^\infty dx_2 \cdots dx_n\, \delta(y+x_2+\cdots+x_n)\, f_\infty(y,x_2,\ldots,x_n) \;+\; \mathcal{O}\big(\Lambda^{-1}\big)\,. \end{aligned} \tag{8}$$

In the previous example the integrand $f_\infty(y, x_2) = 1/(a + x_2)^2$ does not depend on the variable $y$, but in general it could be possible. The important point is that $\Lambda$ disappeared from the integral.

### 2.1 Phase space

Now consider the matrix element $\mathcal{M}\big(\bar{q}, q, \{p_i\}\big)$ involving a quark-antiquark pair with momenta $q^\mu$, $\bar{q}^\mu$ and a number of gluons with momenta $\{p_i^\mu\}$. Let's say the anti-quark is actually an initial-state quark with momentum $q_\Lambda^\mu$ which is light-like, but has components that scale with a parameter $\Lambda$ and become large. The number of gluons is $n + 1$, and this last one is also initial-state.

Consider the phase space integral

$$I = \int d^4q\delta(q^2)\, d\Phi_{1,n}\, \delta^4(q_\Lambda + p_{n+1} - q - p_{1,n})\, \mathcal{M}\big(q_\Lambda, q, \{p_i\}\big)\ , \tag{9}$$

where we abbreviate

$$d\Phi_{1,n} = \prod_{i=1}^{n} d^4p_i\delta(p_i^2) \quad , \quad p_{1,n} = \sum_{i=1}^{n} p_i\ . \tag{10}$$

Now the constraint $\delta^4(q_\Lambda + p_{n+1} - q - p_{1,n})$ demands the large-momentum components of $q_\Lambda^\mu$ to be distributed over the final-state momenta. Let us write the momentum $q_\Lambda^\mu$ explicitly as

$$q_\Lambda^\mu = \Lambda P^\mu\ . \tag{11}$$

In principle, we could give $q_\Lambda^\mu$ finite transverse components plus a vanishing $\bar{P}$-component, but this will not add any value to the procedure. The initial-state gluon is assumed to simply have momentum

$$p_{n+1}^\mu = \bar{x}_m \bar{P}^\mu \tag{12}$$

with some finite $\bar{x}_m$, and no other components. The subscript of $\bar{x}_m$ is chosen to indicate that it is a maximum for the $\bar{P}$-component of the final-state momenta. While not explicitly expressed in the phase space integral, the final-state gluons are implied not to become arbitrarily soft, which implies that the $\bar{P}$ component $\bar{x}_i$ of the momentum of each of them should not vanish. Also, the rapidities $\log(x_i/\bar{x}_i)$ are assumed to be bounded by a given maximum value, which implies that the $P$ component $x_i$ must not become arbitrarily large, and are bounded by some value say $x_m$. So summarizing

$$0 < \bar{x}_i < \bar{x}_m \quad , \quad 0 < x_i < x_m \quad \text{for} \quad i = 1, 2, \ldots n \quad . \tag{13}$$

The idea is to let the final-state quark get the large $P$ component, so at this stage we can say

$$0 < \bar{x}_q < \bar{x}_m \quad , \quad 0 < x_q < \Lambda \quad . \tag{14}$$

We know that $\mathcal{M}\big(q_\Lambda, q, \{p_i\}\big)$ behaves as $\Lambda^2$ if the $P$ component of $q^\mu$ "takes all of $\Lambda$", that is $x_q \to \Lambda - x_k$, and leads to the matrix element with a space like gluon:

$$\mathcal{M}\big(q_\Lambda, q, \{p_i\}\big) \stackrel{\Lambda\to\infty}{\longrightarrow} \Lambda^2\mathcal{M}^*\big(x_kP - q_\perp, \{p_i\}\big) \tag{15}$$

where $\mathcal{M}^*\big(x_kP - q_\perp, \{p_i\}\big)$ is the matrix element with a space-like gluon with momentum $x_kP^\mu - q_\perp^\mu$ instead of the quark-antiquark pair. In order to obtain the correct formula for high-energy factorization, the matrix element still has to be multiplied by a factor $x_k^2|q_\perp|^2/\alpha_s$. In previous work, we always constructed the equivalents of $q_\Lambda^\mu$ and $q^\mu$ such that $q_\Lambda^\mu - q^\mu$ has no $\bar{P}$ component for any value of $\Lambda$, but this is not essential and this component always vanishes for $\Lambda \to \infty$.

In order to provide the large $\Lambda$ component to $x_q$, we introduce an additional integration variable $x_k$ in the phase space integral and we write

$$I = \int dx_k\, d^4q\delta(q^2)\, \delta(x_k + x_q - \Lambda)\, d\Phi_{1,n}\, \delta^4(\Lambda P + p_{n+1} - q - p_{1,n})\, \mathcal{M}\big(\Lambda P, q, \{p_i\}\big)\ . \tag{16}$$

Decomposing $q^\mu$ explicitly:

$$I = \int dx_k \, \frac{dx_q}{2x_q} d\bar{x}_q d^2q_\perp \delta\left(\bar{x}_q - \frac{|q_\perp|^2}{sx_q}\right) \, d\Phi_{1,n} \, \delta(x_k + x_q - \Lambda)$$
$$\times \; \delta^4\big((\Lambda - x_q)P - \bar{x}_q\bar{P} - q_\perp + p_{n+1} - p_{1,n}\big) \, \mathcal{M}\big(\Lambda P \, , \, x_q P + \bar{x}_q \bar{P} + q_\perp \, , \, \{p_i\}\big) \, . \quad (17)$$

Performing the $x_q$-integral eliminating the $\delta(x_k + x_q - \Lambda)$:

$$I = \frac{1}{2} \int dx_k \, d\bar{x}_q d^2q_\perp \delta\left(\bar{x}_q - \frac{|q_\perp|^2}{s(\Lambda - x_k)}\right) \, d\Phi_{1,n}$$
$$\times \; \delta^4\big(x_k P - \bar{x}_q \bar{P} - q_\perp + p_{n+1} - p_{1,n}\big) \, \frac{\mathcal{M}\big(\Lambda P \, , \, (\Lambda - x_k)P + \bar{x}_q \bar{P} + q_\perp \, , \, \{p_i\}\big)}{\Lambda - x_k} \, . \quad (18)$$

Now, we take the limit $\Lambda \to \infty$, implying $\bar{x}_q \to 0$, and we find

$$I = \frac{1}{2} \int dx_k \, d^2q_\perp \, d\Phi_{1,n} \, \delta^4\big(x_k P - q_\perp + p_{n+1} - p_{1,n}\big) \, \frac{\Lambda^2 \mathcal{M}^*\big(x_k P - q_\perp \, , \, \{p_i\}\big)}{\Lambda - x_k} \, , \quad (19)$$

The factor $\Lambda^2/(\Lambda - x_k) \approx \Lambda$ implies that the original "high-energy" phase space integral needs an external factor $1/\Lambda$ in order to be well-defined.

## 2.2 Real radiation

Choosing the correct integration variables to receive the large components was rather simple in the previous phase space integral, because the gluon momenta were simply forbidden to become large. This is different in the phase space integral with real-radiation added:

$$I_R = \int d^4q \delta(q^2) \, d^4r \delta(r^2) \, d\Phi_{1,n} \, \delta^4(\Lambda P + p_{n+1} - q - r - p_{1,n}) \, \mathcal{M}\big(\Lambda P, q, r, \{p_i\}\big) \, . \quad (20)$$

While $r^\mu$ is the momentum of "just" another gluon, it is not restricted as the other ones. In particular, its $P$ component $x_r$ is allowed to become large. This can be expressed in the phase space integral by adding an integration variable $x_k$ again, but then with $\delta(x_k + x_q + x_r - \Lambda)$ intstead of $\delta(x_k + x_q - \Lambda)$. However, naturally one would also want to have a part of phase space where the radiative gluon plays the role of a "normal" gluon regarding the high-energy limit. Thus we introduce a separator $x_{\text{thrs}}$ and write

$$I_R = \int dx_k \, d^4q \delta(q^2) \, d^4r \delta(r^2) \big[\theta(x_{\text{thrs}} - x_r) \; + \theta(x_r - x_{\text{thrs}})\big] \delta(x_k + x_q + x_r - \Lambda)$$
$$\times \, d\Phi_{1,n} \, \delta^4(\Lambda P + p_{n+1} - q - r - p_{1,n}) \, \mathcal{M}\big(\Lambda P, q, r, \{p_i\}\big) \, . \quad (21)$$

The term with $\theta(x_{\rm thrs} - x_r)$ is, regarding the large-$\Lambda$ limit, not different from the Born level integral before, and we will now concentrate on the term with $\theta(x_r - x_{\rm thrs})$.

$$
\begin{aligned}
I_R^{>} = & \int dx_k\, d^4q\delta(q^2)\, d^4r\delta(r^2)\, \theta(x_r - x_{\rm thrs})\, \delta(x_k + x_q + x_r - \Lambda) \\
& \times d\Phi_{1,n}\, \delta^4(\Lambda P + p_{n+1} - q - r - p_{1,n})\, \mathcal{M}\big(\Lambda P, q, r, \{p_i\}\big) \\
= & \int dx_k\, \frac{dx_q}{2x_q} d\bar{x}_q d^2q_\perp \delta\left(\bar{x}_q - \frac{|q_\perp|^2}{s x_q}\right) \frac{dx_r}{2x_r} d\bar{x}_r d^2r_\perp \delta\left(\bar{x}_r - \frac{|r_\perp|^2}{s x_r}\right) \\
& \times \theta(x_r - x_{\rm thrs})\, \delta(x_k + x_q + x_r - \Lambda) \\
& \times d\Phi_{1,n}\, \delta^4(x_k P - q_\perp - r_\perp - (\bar{x}_q + \bar{x}_r)\bar{P} + p_{n+1} - p_{1,n})\, \mathcal{M}\big(\Lambda P, q, r, \{p_i\}\big)\,. \qquad (22)
\end{aligned}
$$

We abbreviate

$$
\Lambda - x_k = \Lambda_k \qquad (23)
$$

and re-scale the variables $x_q$ and $x_r$

$$
\begin{aligned}
I_R^{>} = & \int dx_k \int_0^1 \frac{dx_q}{2x_q} \int d\bar{x}_q d^2q_\perp \delta\left(\bar{x}_q - \frac{|q_\perp|^2}{s x_q \Lambda_k}\right) \int_{x_{\rm thrs}/\Lambda_k}^1 \frac{dx_r}{2x_r} \int d\bar{x}_r d^2r_\perp \delta\left(\bar{x}_r - \frac{|r_\perp|^2}{s x_r \Lambda_k}\right) \\
& \times \Lambda_k^{-1} \delta(x_q + x_r - 1)\, d\Phi_{1,n}\, \delta^4(x_k P - q_\perp - r_\perp - (\bar{x}_q + \bar{x}_r)\bar{P} + p_{n+1} - p_{1,n}) \\
& \times \mathcal{M}\big((\Lambda_k + x_k)P\,,\, x_q \Lambda_k P + q_\perp + \bar{x}_q \bar{P}\,,\, x_r \Lambda_k P + r_\perp + \bar{x}_r \bar{P}\,,\, \{p_i\}\big)\,. \qquad (24)
\end{aligned}
$$

In the following sections, we will show that

$$
\begin{aligned}
& \mathcal{M}\big((\Lambda + x)P\,,\, x_q \Lambda P + q_\perp + \bar{x}_q \bar{P}\,,\, x_r \Lambda P + r_\perp + \bar{x}_r \bar{P}\,,\, \{p_i\}\big) \\
& \qquad \stackrel{\Lambda \to \infty}{\longrightarrow} \mathcal{P}(x_q, q_\perp, x_r, r_\perp)\, \Lambda^2 \mathcal{M}^*\big(xP - q_\perp - r_\perp, \{p_i\}\big)\,, \qquad (25)
\end{aligned}
$$

where

$$
\begin{aligned}
\mathcal{P}(x_q, q_\perp, x_r, r_\perp) = & \Bigg[ C_F\big(x_q |s_{r\bar{q}}| + |s_{qr}|\big) \\
& - \frac{1}{2N_c} \frac{x_q s_{r\bar{q}} - x_r s_{q\bar{q}} - s_{qr}}{\mathrm{sgn}(s_{qr})} \Bigg] \frac{1 + x_q^2}{x_r} \frac{|q_\perp + r_\perp|^2}{|s_{q\bar{q}}||s_{r\bar{q}}||s_{qr}|} \qquad (26)
\end{aligned}
$$

with

$$
s_{q\bar{q}} = \frac{q_\perp^2}{x_q}\ ,\ \ s_{r\bar{q}} = \frac{r_\perp^2}{x_r}\ ,\ \ s_{qr} = (q_\perp + r_\perp)^2 - \frac{q_\perp^2}{x_q} - \frac{r_\perp^2}{x_r}\ . \qquad (27)
$$

So for $\Lambda \to \infty$, we can write the real-radiation integral as

$$
\begin{aligned}
I_R^{>,\rm lim} = & \Lambda \int dx_k d^2k_\perp \mathcal{F}(x_k, k_\perp; x_{\rm thrs}, \Lambda, \bar{x}_m) \\
& \times d\Phi_{1,n}\, \delta^4(x_k P + k_\perp + p_{n+1} - p_{1,n})\, \mathcal{M}^*\big(x_k P + k_\perp, \{p_i\}\big)\,, \qquad (28)
\end{aligned}
$$

where

$$\mathcal{F}(x_k, k_\perp; x_{\rm thrs}, \Lambda, \bar{x}_m) = \int_0^1 \frac{dx_q}{2x_q} \int_0^{\bar{x}_m} d\bar{x}_q \int d^2 q_\perp \delta\left(\bar{x}_q - \frac{|q_\perp|^2}{s x_q \Lambda_k}\right)$$
$$\times \int_{x_{\rm thrs}/\Lambda_k}^1 \frac{dx_r}{2x_r} \int_0^{\bar{x}_m} d\bar{x}_r \int d^2 r_\perp \delta\left(\bar{x}_r - \frac{|r_\perp|^2}{s x_r \Lambda_k}\right)$$
$$\times \delta^2(k_\perp + q_\perp + r_\perp)\, \delta(x_q + x_r - 1)\, \mathcal{P}(x_q, q_\perp, x_r, r_\perp)\,. \quad (29)$$

# 3 High energy limit for amplitudes

Consider the partial amplitude for a case with 2 final-state gluons

$$A(q^-, g_1^-, g_2^+, g_3^+, \bar{q}^+) = \frac{\langle q1\rangle^3 \langle \bar{q}1\rangle}{\langle \bar{q}q\rangle\langle q1\rangle\langle 12\rangle\langle 23\rangle\langle 3\bar{q}\rangle}\,. \quad (30)$$

We imagine the anti-quark $\bar{q}$ to be the initial-state quark with momentum $(\Lambda + x)P^\mu$, so

$$|\bar{q}\rangle = \sqrt{\Lambda + x}\,|P\rangle\,. \quad (31)$$

In case only the momentum of the final-state quark receives the large component, we have

$$|q\rangle = \sqrt{\Lambda}\,|P\rangle + \frac{\kappa^\star}{\sqrt{\Lambda}\,\langle P\bar{P}\rangle}\,|\bar{P}\rangle \quad , \quad |\kappa^\star| = |q_\perp|\,. \quad (32)$$

Contrary to constructions in earlier work, this will give the sum of the quark and antiquark momenta a non-zero component for $\bar{P}^\mu$, which however still vanishes for $\Lambda \to \infty$. Important is that

$$\langle \bar{q}q\rangle = \kappa^\star + \mathcal{O}\big(\Lambda^{-1}\big) \quad (33)$$

does not vanish or become large for $\Lambda \to \infty$, and the amplitude scales as $\Lambda$ due to the occurrence of $|\bar{q}\rangle$ and $|q\rangle$ in other spinor products. Inserting the spinors, we find

$$A(q^-, g_1^-, g_2^+, g_3^+, \bar{q}^+) \to \frac{\Lambda\,\langle P1\rangle^4}{\kappa^\star\langle P1\rangle\langle 12\rangle\langle 23\rangle\langle 3P\rangle} = \Lambda\, A(g^*, g_1^-, g_2^+, g_3^+)\,. \quad (34)$$

Now, we distribute $\Lambda$ over the quark and a gluon:

$$|q\rangle = \sqrt{z\Lambda}\,|P\rangle + \frac{\kappa_q^\star}{\sqrt{z\Lambda}\,\langle P\bar{P}\rangle}\,|\bar{P}\rangle \quad , \quad |\kappa_q^\star| = |q_\perp| \quad (35)$$

$$|r\rangle = \sqrt{(1-z)\Lambda}\,|P\rangle + \frac{\kappa_r^\star}{\sqrt{(1-z)\Lambda}\,\langle P\bar{P}\rangle}\,|\bar{P}\rangle \quad , \quad |\kappa_r^\star| = |r_\perp|\,. \quad (36)$$

This way, $x_q + x_r = \Lambda$ for any $z$, which is imagined to be between 0 and 1. We see that if we choose $r$ to refer to 2 in Eq. (30), then it will produce a factor $\Lambda$ in the denominator and give a subleading contribution. If we choose 1, then the spinor products in the numerator will not

produce powers of $\Lambda$. In order to produce a leading contribution, it must refer to 3, and inserting the spinors, we get

$$\begin{aligned}
A(q^-, g_1^-, g_2^+, g_3^+, \bar{q}^+) &= \frac{\langle q1\rangle^3 \langle \bar{q}1\rangle}{\langle \bar{q}q\rangle\langle q1\rangle\langle 12\rangle\langle 23\rangle\langle 3\bar{q}\rangle} \\
&\to \frac{\left(\sqrt{z\Lambda}\,\langle P1\rangle\right)^3 \left(\sqrt{\Lambda}\,\langle P1\rangle\right)}{\frac{\kappa_q^\star}{\sqrt{z}}\left(\sqrt{z\Lambda}\langle P1\rangle\right)\langle 12\rangle\left(\sqrt{(1-z)\Lambda}\,\langle 2P\rangle\right)\frac{-\kappa_r^\star}{\sqrt{1-z}}} \\
&= \frac{-z^{3/2}\,\Lambda\,\langle P1\rangle^4}{\kappa_q^\star\kappa_r^\star\langle P1\rangle\langle 12\rangle\langle 2P\rangle} = -z^{3/2}\,\frac{\kappa_q^\star+\kappa_r^\star}{\kappa_q^\star\kappa_r^\star}\,\Lambda\,A(g^*, g_1^-, g_2^+)\,, \qquad (37)
\end{aligned}$$

where the factor $\kappa_q^\star + \kappa_r^\star$ occurs, because the space-like gluon in $A(g^*, g_1^-, g_2^+)$ has transverse momentum $q_\perp^\mu + r_\perp^\mu$, and thus the amplitude a factor $\kappa_{q+r}^\star = \kappa_q^\star + \kappa_r^\star$ in the denominator.

## 3.1 Expressions for simple processes

The amplitudes for processes with a single space-like gluon and a pair of on-shell gluons are given by [1]

$$A(g^*, 2^+, 3^-) = \frac{1}{\kappa^\star}\frac{\langle P3\rangle^4}{\langle P2\rangle\langle 23\rangle\langle 3P\rangle} = \frac{1}{\kappa}\frac{[2P]^4}{[P3][32][2P]} \tag{38}$$

$$A(g^*, 2^-, 3^+) = \frac{1}{\kappa^\star}\frac{\langle P2\rangle^4}{\langle P2\rangle\langle 23\rangle\langle 3P\rangle} = \frac{1}{\kappa}\frac{[3P]^4}{[P3][32][2P]} \ . \tag{39}$$

Notice that both the mostly-plus and mostly-minus expressios can be used. Also notice that in the mostly-minus expressions with the square brackets, the ordering in the denominator is backward. In general, to obtain an MHV expression with opposite helicities, we must apply

$$\langle ab\rangle \leftrightarrow [ba] \ . \tag{40}$$

This means that for odd number of spinor products, for example 5,

$$\langle 12\rangle\langle 23\rangle\langle 34\rangle\langle 45\rangle\langle 51\rangle \leftrightarrow -[12][23][34][45][51] \ . \tag{41}$$

## 3.2 Splittings from explicit expressions with auxiliary quarks

There are two kind of splittings: the one for which the radiative gluon "splits from" the quark, denoted $\mathcal{S}$, and the one for which the radiative gluon "splits from" the anti-quark, denoted $\bar{\mathcal{S}}$. Realize that for the square spinor $|\bar{q}]$ there is a minus sign $|\bar{q}] \to -\Lambda|P]$. In the following we denote by $\kappa$ and $\kappa^\star$ the respective quantities related to the transverse momentum of the space-like gluon. So if, for example, the initial-state quark does not have transverse momentum, while the final-state quark and the radiative gluon do, then

$$\kappa = \kappa_q + \kappa_r \quad , \quad \kappa^\star = \kappa^\star_q + \kappa^\star_r \ . \tag{42}$$

To get the splitting $\mathcal{S}(\bar{q}^+, q^-, g_r^-)$, we use

$$\begin{aligned} A(q^-, g_r^-, g_2^-, g_3^+, \bar{q}^+) &= \frac{-[q3][\bar{q}3]^3}{[\bar{q}q][qr][r2][23][3\bar{q}]} \to \frac{z_q^{1/2}}{\sqrt{z_r}[\bar{q}q][qr]}\frac{-\Lambda[P3]^4}{[P2][23][3P]} \\ &= z_q^{1/2}\frac{\kappa}{\sqrt{z_r}[\bar{q}q][qr]}\Lambda A(g^*, g_2^-, g_3^+) \end{aligned} \tag{43}$$

To get the splitting $\mathcal{S}(\bar{q}^+, q^-, g_r^+)$, we use

$$\begin{aligned} A(q^-, g_r^+, g_2^-, g_3^+, \bar{q}^+) &= \frac{\langle q2\rangle^3\langle \bar{q}2\rangle}{\langle \bar{q}q\rangle\langle qr\rangle\langle r2\rangle\langle 23\rangle\langle 3\bar{q}\rangle} \to \frac{z_q^{3/2}}{\sqrt{z_r}\langle \bar{q}q\rangle\langle qr\rangle}\frac{\Lambda\langle P2\rangle^4}{\langle P2\rangle\langle 23\rangle\langle 3P\rangle} \\ &= z_q^{3/2}\frac{\kappa^\star}{\sqrt{z_r}\langle \bar{q}q\rangle\langle qr\rangle}\Lambda A(g^*, g_2^-, g_3^+) \end{aligned} \tag{44}$$

To get the splitting $\mathcal{S}(g_r^-, \bar{q}^+, q^-)$, we use

$$\begin{aligned} A(q^-, g_1^+, g_2^-, g_r^-, \bar{q}^+) &= \frac{-[q1][\bar{q}1]^3}{[\bar{q}q][q1][12][2r][r\bar{q}]} \to \frac{z_q^{1/2}}{\sqrt{z_r z_q}[\bar{q}q][r\bar{q}]} \frac{\Lambda[P1]^4}{[P1][12][2P]} \\ &= -\frac{\kappa}{\sqrt{z_r}[r\bar{q}][\bar{q}q]} \Lambda A(g^*, g_1^+, g_2^-) \end{aligned} \tag{45}$$

To get the splitting $\mathcal{S}(g_r^+, \bar{q}^+, q^-)$, we use

$$\begin{aligned} A(q^-, g_1^+, g_2^-, g_r^+, \bar{q}^+) &= \frac{\langle q2\rangle^3\langle \bar{q}2\rangle}{\langle \bar{q}q\rangle\langle q1\rangle\langle 12\rangle\langle 2r\rangle\langle r\bar{q}\rangle} \to \frac{z_q^{3/2}}{\sqrt{z_r z_q}\langle \bar{q}q\rangle\langle r\bar{q}\rangle} \frac{\Lambda\langle P2\rangle^4}{\langle P1\rangle\langle 12\rangle\langle 2P\rangle} \\ &= z_q \frac{\kappa^\star}{\sqrt{z_r}\langle r\bar{q}\rangle\langle \bar{q}q\rangle} \Lambda A(g^*, g_1^+, g_2^-) \end{aligned} \tag{46}$$

For the opposite helicity of the quark-antiquark pair we get the following.
To get the splitting $\mathcal{S}(\bar{q}^-, q^+, g_r^+)$, we use

$$\begin{aligned} A(q^+, g_r^+, g_2^+, g_3^-, \bar{q}^-) &= \frac{\langle q3\rangle\langle \bar{q}3\rangle^3}{\langle \bar{q}q\rangle\langle qr\rangle\langle r2\rangle\langle 23\rangle\langle 3\bar{q}\rangle} \to \frac{z_q^{1/2}}{\sqrt{z_r}\langle \bar{q}q\rangle\langle qr\rangle} \frac{\Lambda\langle P3\rangle^4}{\langle P2\rangle\langle 23\rangle\langle 3P\rangle} \\ &= z_q^{1/2} \frac{\kappa^\star}{\sqrt{z_r}\langle \bar{q}q\rangle\langle qr\rangle} \Lambda A(g^*, g_2^+, g_3^-) \end{aligned} \tag{47}$$

To get the splitting $\mathcal{S}(\bar{q}^-, q^+, g_r^-)$, we use

$$\begin{aligned} A(q^+, g_r^-, g_2^+, g_3^-, \bar{q}^-) &= \frac{-[q2]^3[\bar{q}2]}{[\bar{q}q][qr][r2][23][3\bar{q}]} \to \frac{z_q^{3/2}}{\sqrt{z_r}[\bar{q}q][qr]} \frac{-\Lambda[P2]^4}{[P2][23][3P]} \\ &= z_q^{3/2} \frac{\kappa}{\sqrt{z_r}[\bar{q}q][qr]} \Lambda A(g^*, g_2^+, g_3^-) \end{aligned} \tag{48}$$

To get the splitting $\mathcal{S}(g_r^+, \bar{q}^-, q^+)$, we use

$$\begin{aligned} A(q^+, g_1^-, g_2^+, g_r^+, \bar{q}^-) &= \frac{\langle q1\rangle\langle \bar{q}1\rangle^3}{\langle \bar{q}q\rangle\langle q1\rangle\langle 12\rangle\langle 2r\rangle\langle r\bar{q}\rangle} \to \frac{\sqrt{z_q}}{\sqrt{z_r z_q}\langle \bar{q}q\rangle\langle r\bar{q}\rangle} \frac{\Lambda\langle P1\rangle^4}{\langle P1\rangle\langle 12\rangle\langle 2P\rangle} \\ &= \frac{\kappa^\star}{\sqrt{z_r}\langle r\bar{q}\rangle\langle \bar{q}q\rangle} \Lambda A(g^*, g_1^-, g_2^+) \end{aligned} \tag{49}$$

To get the splitting $\mathcal{S}(g_r^-, \bar{q}^-, q^+)$, we use

$$\begin{aligned} A(q^+, g_1^-, g_2^+, g_r^-, \bar{q}^-) &= \frac{-[q2]^3[\bar{q}2]}{[\bar{q}q][q1][12][2r][r\bar{q}]} \to \frac{z_q^{3/2}}{\sqrt{z_r z_q}[\bar{q}q][r\bar{q}]} \frac{\Lambda[P2]^4}{[P1][12][2P]} \\ &= -z_q \frac{\kappa}{\sqrt{z_r}[r\bar{q}][\bar{q}q]} \Lambda A(g^*, g_1^-, g_2^+) \end{aligned} \tag{50}$$

Introducing the functions

$$\mathrm{Sqr}(\bar{q}, q, r) = \frac{\kappa}{[\bar{q}q][qr][r\bar{q}]} \quad , \quad \mathrm{Ang}(\bar{q}, q, r) = \frac{\kappa^\star}{\langle \bar{q}q\rangle\langle qr\rangle\langle r\bar{q}\rangle} \, , \tag{51}$$

we summarize

$$\mathcal{S}(\bar{q}^+, q^-, g_r^-) = \mathrm{Sqr}(\bar{q}, q, r)\,\frac{z_q^{1/2}[r\bar{q}]}{\sqrt{z_r}} \quad , \quad \mathcal{S}(\bar{q}^-, q^+, g_r^+) = \mathrm{Ang}(\bar{q}, q, r)\,\frac{z_q^{1/2}\langle r\bar{q}\rangle}{\sqrt{z_r}} \quad , \tag{52}$$

$$\mathcal{S}(\bar{q}^+, q^-, g_r^+) = \mathrm{Ang}(\bar{q}, q, r)\,\frac{z_q^{3/2}\langle r\bar{q}\rangle}{\sqrt{z_r}} \quad , \quad \mathcal{S}(\bar{q}^-, q^+, g_r^-) = \mathrm{Sqr}(\bar{q}, q, r)\,\frac{z_q^{3/2}[r\bar{q}]}{\sqrt{z_r}} \quad , \tag{53}$$

$$\mathcal{S}(g_r^-, \bar{q}^+, q^-) = \mathrm{Sqr}(\bar{q}, q, r)\,\frac{-[qr]}{\sqrt{z_r}} \quad , \quad \mathcal{S}(g_r^+, \bar{q}^-, q^+) = \mathrm{Ang}(\bar{q}, q, r)\,\frac{\langle qr\rangle}{\sqrt{z_r}} \quad , \tag{54}$$

$$\mathcal{S}(g_r^+, \bar{q}^+, q^-) = \mathrm{Ang}(\bar{q}, q, r)\,\frac{z_q\langle qr\rangle}{\sqrt{z_r}} \quad , \quad \mathcal{S}(g_r^-, \bar{q}^-, q^+) = \mathrm{Sqr}(\bar{q}, q, r)\,\frac{-z_q[qr]}{\sqrt{z_r}} \quad . \tag{55}$$

Let us introduct the short-hand notation

$$\mathcal{S}_\lambda^h = \mathcal{S}(\bar{q}^\lambda, q^{-\lambda}, g_r^h) \quad , \quad \bar{\mathcal{S}}_\lambda^h = \mathcal{S}(g_r^h, \bar{q}^\lambda, q^{-\lambda}) \,. \tag{56}$$

To calculate the squared splitting function, we will only use the relations

$$\left|\mathcal{S}_\lambda^h\right| = \left|\mathcal{S}_{-\lambda}^{-h}\right| \quad , \quad \left|\bar{\mathcal{S}}_\lambda^h\right| = \left|\bar{\mathcal{S}}_{-\lambda}^{-h}\right| \tag{57}$$

and

$$\left|\mathcal{S}_h^h\right| = z_q\left|\mathcal{S}_h^{-h}\right| \quad , \quad \left|\bar{\mathcal{S}}_h^h\right| = z_q\left|\bar{\mathcal{S}}_h^{-h}\right| \tag{58}$$

and

$$\mathcal{S}_h^\pm = \pm\sqrt{z_q}\,\frac{(r\bar{q})_\mp}{(qr)_\mp}\,\bar{\mathcal{S}}_h^\pm \tag{59}$$

with

$$(ab)_- = \langle ab\rangle \quad , \quad (ab)_+ = [ab] \,, \tag{60}$$

and

$$\left|\bar{\mathcal{S}}_\lambda^{-\lambda}\right|^2 = \frac{|k_\perp|^2}{z_r|s_{q\bar{q}}||s_{r\bar{q}}|} \,. \tag{61}$$

## 3.3 Squared amplitudes with auxiliary quarks

In general an amplitude for $(n+1)$ gluons and a quark-antiquark pair can for any given helicity configuration be decomposed as

$$\mathcal{A}_{ji}^{a_1 a_2 \cdots a_{n+1}} = \sum_{\sigma\in S_{n+1}} \left(T^{a_{\sigma(1)}} T^{a_{\sigma(2)}} \cdots T^{a_{\sigma(n+1)}}\right)_{ji} A(q, g_{\sigma(1)}, g_{\sigma(2)}, \ldots, g_{\sigma(n+1)}, \bar{q}) \tag{62}$$

where $A$ without color indices is the partial amplitude. Now, gluon number $n+1$ is the radiative one, and only when it is the first or the last in the argument list, then the partial amplitude contributes in the joint large-$\Lambda$ limit for the radiative gluon and the auxiliary pair:

$$\Lambda^{-1}\mathcal{A}_{ji}^{a_1 a_2 \cdots a_n a_r} \to \left(T^{a_r}\Omega\right)_{ji}\mathcal{S} + \left(\Omega T^{a_r}\right)_{ji}\bar{\mathcal{S}} \,, \tag{63}$$

where

$$\Omega_{lk} = \sum_{\sigma\in S_n} \left(T^{a_{\sigma(1)}} T^{a_{\sigma(2)}} \cdots T^{a_{\sigma(n)}}\right)_{lk} A(g^*, g_{\sigma(1)}, g_{\sigma(2)}, \ldots, g_{\sigma(n)}) \tag{64}$$

is the full colored amplitude with $1$ space-like and $n$ on-shell gluons. The square of the amplitude summed over the color degrees of freedom of the the space-like gluon is given by

$$\Omega_{kl}\Omega^\star_{kl} = \mathrm{Tr}\big(\Omega\Omega^\dagger\big) \ . \tag{65}$$

Realize that the fundamental indices $j, i$ are for the space-like gluon, and indeed

$$\mathrm{Tr}(\Omega) = 0 \tag{66}$$

thanks to the eikonal quark line (see appendix C in [2]), so it has only $N_c^2-1$ degrees of freedom. One could give the space-like gluon a true adjoint color index $a_*$ by contracting with $(T^{a_*})_{kl}$. Squaring and summing only over the color of the radiative gluon and the space-like gluon, we get

$$\begin{aligned}
&\left(\mathcal{S}T^{a_r}_{jk}\Omega_{ki} + \bar{\mathcal{S}}T^{a_r}_{ki}\Omega_{jk}\right)\left(\mathcal{S}T^{a_r}_{jl}\Omega_{li} + \bar{\mathcal{S}}T^{a_r}_{li}\Omega_{jl}\right)^\star \\
&\quad = |\mathcal{S}|^2 T^{a_r}_{jk}T^{a_r}_{lj}\Omega_{ki}\Omega^\dagger_{il} + |\bar{\mathcal{S}}|^2 T^{a_r}_{ki}T^{a_r}_{il}\Omega_{jk}\Omega^\dagger_{lj} + \mathcal{S}\bar{\mathcal{S}}^\star T^{a_r}_{jk}T^{a_r}_{il}\Omega_{ki}\Omega^\dagger_{lj} + \mathcal{S}^\star\bar{\mathcal{S}}T^{a_r}_{ki}T^{a_r}_{lj}\Omega_{jk}\Omega^\dagger_{il} \\
&\quad = |\mathcal{S}|^2\mathrm{Tr}\big(T^{a_r}\Omega\Omega^\dagger T^{a_r}\big) + |\bar{\mathcal{S}}|^2\mathrm{Tr}\big(\Omega T^{a_r}T^{a_r}\Omega^\dagger\big) + \mathcal{S}\bar{\mathcal{S}}^\star\mathrm{Tr}\big(T^{a_r}\Omega T^{a_r}\Omega^\dagger\big) + \mathcal{S}^\star\bar{\mathcal{S}}\mathrm{Tr}\big(\Omega T^{a_r}\Omega^\dagger T^{a_r}\big) \\
&\quad = \left[C_F|\mathcal{S}|^2 + C_F|\bar{\mathcal{S}}|^2 - \tfrac{1}{2N_c}\big(\mathcal{S}\bar{\mathcal{S}}^\star + \mathcal{S}^\star\bar{\mathcal{S}}\big)\right]\mathrm{Tr}\big(\Omega\Omega^\dagger\big) \ ,
\end{aligned} \tag{67}$$

where we used that $\mathrm{Tr}(\Omega) = 0$. In order to evaluate the $1/N_c$-term further, we look at specific helicities for the radiative gluon and the auxiliary partons. Remember that we have

$$\mathcal{S}^\mp_\lambda = \mp\sqrt{z_q}\,\frac{(r\bar{q})_\pm}{(qr)_\pm}\,\bar{\mathcal{S}}^\mp_\lambda \ . \tag{68}$$

Now, because $\bar{q}$ is initial-state while $q$ and $r$ are final-state, we have

$$(r\bar{q})^\star_\pm = (r\bar{q})_\mp \quad \text{while} \quad (qr)^\star_\pm = -(qr)_\mp \quad , \tag{69}$$

so

$$\begin{aligned}
\mathcal{S}^{\mp\star}_\lambda\bar{\mathcal{S}}^\mp_\lambda + \mathcal{S}^\mp_\lambda\bar{\mathcal{S}}^{\mp\star}_\lambda &= \mp\sqrt{z_q}|\bar{\mathcal{S}}^\mp_\lambda|^2\left(\frac{(r\bar{q})^\star_\pm}{(qr)^\star_\pm} + \frac{(r\bar{q})_\pm}{(qr)_\pm}\right) = \mp\sqrt{z_q}|\bar{\mathcal{S}}^\mp_\lambda|^2\left(\frac{(r\bar{q})_\pm}{(qr)_\pm} - \frac{(r\bar{q})_\mp}{(qr)_\mp}\right) \\
&= \mp\sqrt{z_q}|\bar{\mathcal{S}}^\mp_\lambda|^2\frac{(r\bar{q})_\pm(qr)_\mp - (r\bar{q})_\mp(qr)_\pm}{-s_{rq}} \\
&= \sqrt{z_q}|\bar{\mathcal{S}}^\mp_\lambda|^2\frac{[r\bar{q}]\langle qr\rangle - \langle r\bar{q}\rangle[qr]}{s_{rq}} \ .
\end{aligned} \tag{70}$$

Using

$$|\bar{\mathcal{S}}^\lambda_\lambda| = z_q|\bar{\mathcal{S}}^{-\lambda}_\lambda| \tag{71}$$

we find

$$\sum_{h=\pm}\left[\mathcal{S}_\lambda^{h\star}\bar{\mathcal{S}}_\lambda^h+\mathcal{S}_\lambda^h\bar{\mathcal{S}}_\lambda^{h\star}\right]=\sqrt{z_q}(1+z_q^2)|\bar{\mathcal{S}}_\lambda^{-\lambda}|^2\frac{[r\bar{q}]\langle qr\rangle-\langle r\bar{q}\rangle[qr]}{s_{rq}}\ . \tag{72}$$

Since $|\bar{\mathcal{S}}_+^-|=|\bar{\mathcal{S}}_-^+|$, we see that the two contributions for $\lambda=\pm$ are identical. Using the explicit expressions for the spinor products (154), we find

$$[r\bar{q}]\langle qr\rangle-\langle r\bar{q}\rangle[qr]=\frac{2}{\sqrt{z_q}}(\vec{q}_\perp\cdot\vec{r}_\perp)-2\frac{\sqrt{z_q}}{z_r}|r_\perp|^2=\frac{2}{\sqrt{z_q}\,z_r}\left[z_r(\vec{q}_\perp\cdot\vec{r}_\perp)-z_q|r_\perp|^2\right]\ . \tag{73}$$

We put the arrows above the transverse momenta to indicate that there is no minus sign coming from the inner product. A more general formula, in case also $\bar{q}$ has non-vanishing transverse momentum, is

$$[r\bar{q}]\langle qr\rangle-\langle r\bar{q}\rangle[qr]=\frac{1}{\sqrt{z_q}}\left(z_q s_{r\bar{q}}-z_r s_{q\bar{q}}-s_{rq}\right)\ . \tag{74}$$

Combining:

$$\begin{aligned}\sum_{h=\pm}\left[\mathcal{S}_\lambda^{h\star}\bar{\mathcal{S}}_\lambda^h+\mathcal{S}_\lambda^h\bar{\mathcal{S}}_\lambda^{h\star}\right]&=(1+z_q^2)\,\frac{z_q s_{r\bar{q}}-z_r s_{q\bar{q}}-s_{rq}}{\mathrm{sgn}(s_{rq})|s_{rq}|}\,|\bar{\mathcal{S}}_\lambda^{-\lambda}|^2\\&=\frac{1+z_q^2}{z_r}\,\frac{z_q s_{r\bar{q}}-z_r s_{q\bar{q}}-s_{rq}}{\mathrm{sgn}(s_{rq})}\,\frac{|\bar{q}_\perp+q_\perp+r_\perp|^2}{|s_{q\bar{q}}||s_{r\bar{q}}||s_{rq}|}\ .\end{aligned} \tag{75}$$

For the "leading color" contribution, we find

$$\begin{aligned}\sum_{h=\pm}\left[|\mathcal{S}_\lambda^h|^2+|\bar{\mathcal{S}}_\lambda^h|^2\right]&=\sum_{h=\pm}\left(1+z_q\frac{|s_{r\bar{q}}|}{|s_{qr}|}\right)|\bar{\mathcal{S}}_\lambda^h|^2=(1+z_q^2)\left(1+z_q\frac{|s_{r\bar{q}}|}{|s_{qr}|}\right)|\bar{\mathcal{S}}_\lambda^{-\lambda}|^2\\&=\frac{1+z_q^2}{z_r}\left(|s_{qr}|+z_q|s_{r\bar{q}}|\right)\frac{|\bar{q}_\perp+q_\perp+r_\perp|^2}{|s_{qr}||s_{q\bar{q}}||s_{r\bar{q}}|}\ .\end{aligned} \tag{76}$$

Here we wrote the general formulas valid also if $\bar{q}_\perp$ is non-vanishing.

## 3.4 Splittings from explicit expressions with auxiliary gluons

We label the two gluons as $g_q$ and $g_{\bar{q}}$ to keep the notation close to the previous section. We must realize, however, that the two final-state gluons labelled $g_r$ and $g_q$ are equivalent, and we need to consider more orderings and helicity configurations than in the case of auxiliary quarks. We will only show the cases in which $g_{\bar{q}}$ has positive helicity. The other ones are obtained via $\langle ab\rangle\leftrightarrow[ba]$ and $\kappa\leftrightarrow\kappa^\star$.

To get the splitting $\mathcal{S}(g_{\bar{q}}^+,g_q^-,g_r^-)$, we use

$$\begin{aligned}A(g_q^-,g_r^-,g_2^-,g_3^+,g_{\bar{q}}^+)&=\frac{-[\bar{q}3]^4}{[\bar{q}q][qr][r2][23][3\bar{q}]}\longrightarrow\frac{1}{\sqrt{z_r}[\bar{q}q][qr]}\frac{\Lambda[P3]^4}{[P2][23][3P]}\\&=-\frac{\kappa}{\sqrt{z_r}[\bar{q}q][qr]}\,\Lambda A(g^*,g_2^-,g_3^+)\end{aligned} \tag{77}$$

To get the splitting $\mathcal{S}(g^+_{\bar{q}}, g^-_q, g^+_r)$, we use

$$\begin{aligned}\mathcal{A}(g^-_q, g^+_r, g^-_2, g^+_3, g^+_{\bar{q}}) &= \frac{\langle q2\rangle^4}{\langle \bar{q}q\rangle\langle qr\rangle\langle r2\rangle\langle 23\rangle\langle 3\bar{q}\rangle} \to \frac{z_q^2}{\sqrt{z_r}\langle \bar{q}q\rangle\langle qr\rangle}\frac{\Lambda\langle P2\rangle^4}{\langle P2\rangle\langle 23\rangle\langle 3P\rangle} \\ &= z_q^2 \frac{\kappa^\star}{\sqrt{z_r}\langle \bar{q}q\rangle\langle qr\rangle}\,\Lambda\mathcal{A}(g^*, g^-_2, g^+_3) \end{aligned} \tag{78}$$

Now we also need to determine $\mathcal{S}(g^+_{\bar{q}}, g^+_q, g^-_r)$, for example from

$$\begin{aligned}\mathcal{A}(g^+_q, g^-_r, g^+_2, g^-_3, g^+_{\bar{q}}) &= \frac{\langle r3\rangle^4}{\langle \bar{q}q\rangle\langle qr\rangle\langle r2\rangle\langle 23\rangle\langle 3\bar{q}\rangle} \to \frac{z_r^2}{\sqrt{z_r}\langle \bar{q}q\rangle\langle qr\rangle}\frac{\Lambda\langle P3\rangle^4}{\langle P2\rangle\langle 23\rangle\langle 3P\rangle} \\ &= z_r^2 \frac{\kappa^\star}{\sqrt{z_r}\langle \bar{q}q\rangle\langle qr\rangle}\,\Lambda\mathcal{A}(g^*, g^+_2, g^-_3) \end{aligned} \tag{79}$$

$$\begin{aligned}\mathcal{A}(g^-_q, g^+_1, g^-_2, g^-_r, g^+_{\bar{q}}) &= \frac{-[\bar{q}1]^4}{[\bar{q}q][q1][12][2r][r\bar{q}]} \to \frac{1}{\sqrt{z_r z_q}[\bar{q}q][r\bar{q}]}\frac{-\Lambda[P1]^4}{[P1][12][2P]} \\ &= z_q^{-1/2} \frac{\kappa}{\sqrt{z_r}[r\bar{q}][\bar{q}q]}\,\Lambda\mathcal{A}(g^*, g^+_1, g^-_2) \end{aligned} \tag{80}$$

To get the splitting $\mathcal{S}(g^+_r, g^+_{\bar{q}}, g^-_q)$, we use

$$\begin{aligned}\mathcal{A}(g^-_q, g^+_1, g^-_2, g^+_r, g^+_{\bar{q}}) &= \frac{\langle q2\rangle^4}{\langle \bar{q}q\rangle\langle q1\rangle\langle 12\rangle\langle 2r\rangle\langle r\bar{q}\rangle} \to \frac{z_q^2}{\sqrt{z_r z_q}\langle \bar{q}q\rangle\langle r\bar{q}\rangle}\frac{\Lambda\langle P2\rangle^4}{\langle P1\rangle\langle 12\rangle\langle 2P\rangle} \\ &= z_q^{3/2} \frac{\kappa^\star}{\sqrt{z_r}\langle r\bar{q}\rangle\langle \bar{q}q\rangle}\,\Lambda\mathcal{A}(g^*, g^+_1, g^-_2) \end{aligned} \tag{81}$$

Now, we also need to obtain $\mathcal{S}(g^-_r, g^+_{\bar{q}}, g^+_q)$, for exmple from

$$\begin{aligned}\mathcal{A}(g^+_q, g^-_1, g^+_2, g^-_r, g^+_{\bar{q}}) &= \frac{\langle r1\rangle^4}{\langle \bar{q}q\rangle\langle q1\rangle\langle 12\rangle\langle 2r\rangle\langle r\bar{q}\rangle} \to \frac{z_r^2}{\sqrt{z_r z_q}\langle \bar{q}q\rangle\langle r\bar{q}\rangle}\frac{\Lambda\langle P1\rangle^4}{\langle P1\rangle\langle 12\rangle\langle 2P\rangle} \\ &= z_r^{3/2} \frac{\kappa^\star}{\sqrt{z_q}\langle r\bar{q}\rangle\langle \bar{q}q\rangle}\,\Lambda\mathcal{A}(g^*, g^-_1, g^+_2) \end{aligned} \tag{82}$$

Finally, we can get the splittings $\mathcal{S}(g_q, g_r, g^+_{\bar{q}})$ for example from the ones $\mathcal{S}(g^+_{\bar{q}}, g_q, g_r)$ above by exchanging $r \leftrightarrow q$. The order of the arguments is chosen such that the one whos spinor appears twice in the denominator is in the middle. The function are symmetric in exchangeing the spinors of the outer arguments. Rather than expressing relations between the functions, we

just summarize

$$\mathcal{S}(g^+_{\bar{q}}, g^-_q, g^-_r) = \mathrm{Sqr}(\bar{q}, q, r)\, \frac{-z_q^{1/2}[r\bar{q}]}{\sqrt{z_q z_r}} \tag{83}$$

$$\mathcal{S}(g^+_{\bar{q}}, g^-_q, g^+_r) = \mathrm{Ang}(\bar{q}, q, r)\, \frac{z_q^{5/2}\langle r\bar{q}\rangle}{\sqrt{z_q z_r}} \tag{84}$$

$$\mathcal{S}(g^+_{\bar{q}}, g^+_q, g^-_r) = \mathrm{Ang}(\bar{q}, q, r)\, \frac{z_q^{1/2} z_r^2\langle r\bar{q}\rangle}{\sqrt{z_q z_r}} \tag{85}$$

$$\mathcal{S}(g^-_r, g^+_{\bar{q}}, g^-_q) = \mathrm{Sqr}(\bar{q}, q, r)\, \frac{[qr]}{\sqrt{z_q z_r}} \tag{86}$$

$$\mathcal{S}(g^+_r, g^+_{\bar{q}}, g^-_q) = \mathrm{Ang}(\bar{q}, q, r)\, \frac{z_q^2\langle qr\rangle}{\sqrt{z_q z_r}} \tag{87}$$

$$\mathcal{S}(g^-_r, g^+_{\bar{q}}, g^+_q) = \mathrm{Ang}(\bar{q}, q, r)\, \frac{z_r^2\langle qr\rangle}{\sqrt{z_q z_r}} \tag{88}$$

$$\mathcal{S}(g^-_q, g^-_r, g^+_{\bar{q}}) = \mathrm{Sqr}(\bar{q}, q, r)\, \frac{-z_r^{1/2}[q\bar{q}]}{\sqrt{z_q z_r}} \tag{89}$$

$$\mathcal{S}(g^+_q, g^-_r, g^+_{\bar{q}}) = \mathrm{Ang}(\bar{q}, q, r)\, \frac{z_r^{5/2}\langle q\bar{q}\rangle}{\sqrt{z_q z_r}} \tag{90}$$

$$\mathcal{S}(g^-_q, g^+_r, g^+_{\bar{q}}) = \mathrm{Ang}(\bar{q}, q, r)\, \frac{z_r^{1/2} z_q^2\langle q\bar{q}\rangle}{\sqrt{z_q z_r}} \tag{91}$$

### 3.5 Squared amplitudes with auxiliary gluons

In order to obtain an amplitude for 1 space-like gluon and $n$ on-shell gluons using auxiliary gluons, a subset of $(n+2)$-gluon partial amplitudes must be selected, namely the ones for which the auxiliary gluons are next to each other. For the fully on-shell amplitude, we have

$$\mathcal{A}^{a_1\cdots a_{n+1}a_{n+2}} = \sum_{\sigma\in S_{n+1}} \left(T^{a_{\sigma(1)}}\cdots T^{a_{\sigma(n+1)}}T^{a_{n+2}}\right) A(g_{\sigma(1)},\ldots,g_{\sigma(n+1)},g_{n+2})\,, \tag{92}$$

while to obtain the 1-space-like amplitude, we let gluons number $(n+1, n+2)$ take the auxiliary role and give them labels $(\bar{q}, q)$, and we only need

$$\sum_{\sigma\in S_n} \left(T^{a_{\sigma(1)}}\cdots T^{a_{\sigma(n)}}T^{a_{\bar{q}}}T^{a_q}\right) A(g_{\sigma(1)},\ldots,g_{\sigma(n)},g_{\bar{q}},g_q)\,. \tag{93}$$

We are allowed to make such a selection, since the partial amplitudes are gauge invariant by themselves. In fact, it is easy to see that partial amplitudes for which the auxiliary pair is not adjacent are subleading in the large-$\Lambda$ limit. For the partial amplitudes in general we may permute arguments cyclicly, and we have

$$A(g_{\sigma(1)},\ldots,g_{\sigma(n)},g_{\bar{q}},g_q) = A(g_q,g_{\sigma(1)},\ldots,g_{\sigma(n)},g_{\bar{q}}) \to \Lambda A(g^*,g_{\sigma(1)},\ldots,g_{\sigma(n)})\,. \tag{94}$$

The pair $T^{a_{\bar q}}T^{a_q}$ can be replaced with $T^{a_*}$ for the space-like gluon, or alternatively, contracted with $(-2i/N_c)f^{a^* a_{\bar q} a_q}$. We however prefer to keep both generators explicitly now. Applying cyclic permutation symmetry also for the trace, we get

$$\mathcal{A}_*^{a_1\cdots a_n a_{\bar q} a_q} = \sum_{\sigma\in S_n} \left(T^{q_{\bar q}}T^{a_q}T^{a_{\sigma(1)}}\cdots T^{a_{\sigma(n)}}\right) A(g^*, g_{\sigma(1)},\ldots,g_{\sigma(n)})\,. \tag{95}$$

To make contact with Section 3.3 we note that we can write

$$\mathcal{A}_*^{a_1\cdots a_n a_{\bar q} a_q} = \mathrm{Tr}\big(T^{a_{\bar q}}T^{a_q}\Omega\big)\,, \tag{96}$$

with $\Omega$ from Eq. (64), and we remind the reader that $\mathrm{Tr}(\Omega) = 0$, so the square summed over $a_{\bar q}, a_q$ is

$$\sum_{a_{\bar q},a_q} |\mathcal{A}_*^{a_1\cdots a_n a_{\bar q} a_q}| = \mathrm{Tr}\big(T^{a_{\bar q}}T^{a_q}\Omega\big)\mathrm{Tr}\big(\Omega^\dagger T^{a_q}T^{a_{\bar q}}\big) = \frac{N_c^2-2}{4N_c}\,\mathrm{Tr}\big(\Omega\Omega^\dagger\big)\,. \tag{97}$$

In the joint limit of the auxiliary pair and an extra radiative gluon, only partial amplitudes survive in which all three of them are adjacent. Within the freedom of cyclic symmetry, we can write

$$\begin{aligned}\Lambda^{-1}\mathcal{A}^{a_1\cdots a_n a_r a_{\bar q} a_q} \to\ & \mathrm{Tr}\big(T^{a_{\bar q}}T^{a_q}T^{a_r}\Omega\big)\,\mathcal{S}(g_{\bar q},g_q,g_r) + \mathrm{Tr}\big(T^{a_r}T^{a_{\bar q}}T^{a_q}\Omega\big)\,\mathcal{S}(g_r,g_{\bar q},g_q)\\ & + \mathrm{Tr}\big(T^{a_q}T^{a_r}T^{a_{\bar q}}\Omega\big)\,\mathcal{S}(g_q,g_r,g_{\bar q})\,.\end{aligned} \tag{98}$$

For the square, summed over the color of $g_{\bar q}, g_q, g_r$ and summed over the helicities of $g_q, g_r$, we find

$$\begin{aligned}&\sum_{h_q,h_r}\sum_{a_{\bar q},a_q,a_r} \left|\Lambda^{-1}\mathcal{A}^{a_1\cdots a_n a_r a_{\bar q} a_q}\right|^2\\ &\quad\to \frac{N_c^2-2}{8}\,\mathrm{Tr}\big(\Omega\Omega^\dagger\big)\,\frac{1+z_q^4+z_r^4}{z_q z_r}\,\frac{|\bar q_\perp+q_\perp+r_\perp|^2}{|s_{qr}||s_{q\bar q}||s_{r\bar q}|}\big(|s_{rq}|+z_q|s_{r\bar q}|+z_r|s_{q\bar q}|\big)\,.\end{aligned} \tag{99}$$

Here we used that $s_{q\bar q}<0$ and $s_{r\bar q}<0$ while $s_{rq}>0$. Summing over the helicity of $g_{\bar q}$ will give an extra factor 2.

## 3.6 Splittings for initial-state gluon and final-state quark-antiquark pair

We will still use the label $r$ for the gluon, but realize that now for $|r]$ there is a minus sign $|r] \to -\Lambda|P]$, and a momenum fraction $z_{\bar q}$ will be associated with the final-state anti-quark.

To get the splitting $\mathcal{S}_g(\bar q^+, q^-, g_r^-)$, we use

$$\begin{aligned}A(q^-,g_r^-,g_2^-,g_3^+,\bar q^+) &= \frac{-[q3][\bar q3]^3}{[\bar q q][qr][r2][23][3\bar q]} \to \frac{z_q^{1/2}z_{\bar q}}{[\bar q q][qr]}\,\frac{\Lambda[P3]^4}{[P2][23][3P]}\\ &= -z_q^{1/2}z_{\bar q}\,\frac{\kappa}{[\bar q q][qr]}\,\Lambda A(g^*,g_2^-,g_3^+)\end{aligned} \tag{100}$$

To get the splitting $\mathcal{S}_g(\bar{q}^+, q^-, g_r^+)$, we use

$$\begin{aligned} A(q^-, g_r^+, g_2^-, g_3^+, \bar{q}^+) &= \frac{\langle q2\rangle^3\langle \bar{q}2\rangle}{\langle \bar{q}q\rangle\langle qr\rangle\langle r2\rangle\langle 23\rangle\langle 3\bar{q}\rangle} \to \frac{z_q^{3/2}}{\langle \bar{q}q\rangle\langle qr\rangle}\frac{\Lambda\langle P2\rangle^4}{\langle P2\rangle\langle 23\rangle\langle 3P\rangle} \\ &= z_q^{3/2}\frac{\kappa^\star}{\langle \bar{q}q\rangle\langle qr\rangle}\,\Lambda A(g^*, g_2^-, g_3^+) \end{aligned} \tag{101}$$

To get the splitting $\mathcal{S}_g(g_r^-, \bar{q}^+, q^-)$, we use

$$\begin{aligned} A(q^-, g_1^+, g_2^-, g_r^-, \bar{q}^+) &= \frac{-[q1][\bar{q}1]^3}{[\bar{q}q][q1][12][2r][r\bar{q}]} \to \frac{z_{\bar{q}}^{3/2}}{[\bar{q}q][r\bar{q}]}\frac{\Lambda[P1]^4}{[P1][12][2P]} \\ &= -z_{\bar{q}}^{3/2}\frac{\kappa}{[r\bar{q}][\bar{q}q]}\,\Lambda A(g^*, g_1^+, g_2^-) \end{aligned} \tag{102}$$

To get the splitting $\mathcal{S}_g(g_r^+, \bar{q}^+, q^-)$, we use

$$\begin{aligned} A(q^-, g_1^+, g_2^-, g_r^+, \bar{q}^+) &= \frac{\langle q2\rangle^3\langle \bar{q}2\rangle}{\langle \bar{q}q\rangle\langle q1\rangle\langle 12\rangle\langle 2r\rangle\langle r\bar{q}\rangle} \to \frac{z_q z_{\bar{q}}^{1/2}}{\langle \bar{q}q\rangle\langle r\bar{q}\rangle}\frac{\Lambda\langle P2\rangle^4}{\langle P1\rangle\langle 12\rangle\langle 2P\rangle} \\ &= z_q z_{\bar{q}}^{1/2}\frac{\kappa^\star}{\langle r\bar{q}\rangle\langle \bar{q}q\rangle}\,\Lambda A(g^*, g_1^+, g_2^-) \end{aligned} \tag{103}$$

For the opposite helicity of the quark-antiquark pair we get the following.
To get the splitting $\mathcal{S}_g(\bar{q}^-, q^+, g_r^+)$, we use

$$\begin{aligned} A(q^+, g_r^+, g_2^+, g_3^-, \bar{q}^-) &= \frac{\langle q3\rangle\langle \bar{q}3\rangle^3}{\langle \bar{q}q\rangle\langle qr\rangle\langle r2\rangle\langle 23\rangle\langle 3\bar{q}\rangle} \to \frac{z_q^{1/2} z_{\bar{q}}}{\langle \bar{q}q\rangle\langle qr\rangle}\frac{\Lambda\langle P3\rangle^4}{\langle P2\rangle\langle 23\rangle\langle 3P\rangle} \\ &= z_q^{1/2} z_{\bar{q}}\frac{\kappa^\star}{\langle \bar{q}q\rangle\langle qr\rangle}\,\Lambda A(g^*, g_2^+, g_3^-) \end{aligned} \tag{104}$$

To get the splitting $\mathcal{S}_g(\bar{q}^-, q^+, g_r^-)$, we use

$$\begin{aligned} A(q^+, g_r^-, g_2^+, g_3^-, \bar{q}^-) &= \frac{-[q2]^3[\bar{q}2]}{[\bar{q}q][qr][r2][23][3\bar{q}]} \to \frac{z_q^{3/2}}{[\bar{q}q][qr]}\frac{\Lambda[P2]^4}{[P2][23][3P]} \\ &= -z_q^{3/2}\frac{\kappa}{[\bar{q}q][qr]}\,\Lambda A(g^*, g_2^+, g_3^-) \end{aligned} \tag{105}$$

To get the splitting $\mathcal{S}_g(g_r^+, \bar{q}^-, q^+)$, we use

$$\begin{aligned} A(q^+, g_1^-, g_2^+, g_r^+, \bar{q}^-) &= \frac{\langle q1\rangle\langle \bar{q}1\rangle^3}{\langle \bar{q}q\rangle\langle q1\rangle\langle 12\rangle\langle 2r\rangle\langle r\bar{q}\rangle} \to \frac{z_{\bar{q}}^{3/2}}{\langle \bar{q}q\rangle\langle r\bar{q}\rangle}\frac{\Lambda\langle P1\rangle^4}{\langle P1\rangle\langle 12\rangle\langle 2P\rangle} \\ &= z_{\bar{q}}^{3/2}\frac{\kappa^\star}{\langle r\bar{q}\rangle\langle \bar{q}q\rangle}\,\Lambda A(g^*, g_1^-, g_2^+) \end{aligned} \tag{106}$$

To get the splitting $\mathcal{S}_g(g_r^-, \bar{q}^-, q^+)$, we use

$$\begin{aligned} A(q^+, g_1^-, g_2^+, g_r^-, \bar{q}^-) &= \frac{-[q2]^3[\bar{q}2]}{[\bar{q}q][q1][12][2r][r\bar{q}]} \to \frac{z_q z_{\bar{q}}^{1/2}}{[\bar{q}q][r\bar{q}]}\frac{\Lambda[P2]^4}{[P1][12][2P]} \\ &= -z_q z_{\bar{q}}^{1/2}\frac{\kappa}{[r\bar{q}][\bar{q}q]}\,\Lambda A(g^*, g_1^-, g_2^+) \end{aligned} \tag{107}$$

We summarize

$$\mathcal{S}_g(\bar{q}^+, q^-, g_r^-) = \mathrm{Sqr}(\bar{q}, q, r)\,\big(-z_q^{1/2} z_{\bar{q}}[r\bar{q}]\big) \tag{108}$$
$$\mathcal{S}_g(\bar{q}^+, q^-, g_r^+) = \mathrm{Ang}(\bar{q}, q, r)\,\big(z_q^{3/2}\langle r\bar{q}\rangle\big) \tag{109}$$
$$\mathcal{S}_g(g_r^-, \bar{q}^+, q^-) = \mathrm{Sqr}(\bar{q}, q, r)\,\big(-z_{\bar{q}}^{3/2}[qr]\big) \tag{110}$$
$$\mathcal{S}_g(g_r^+, \bar{q}^+, q^-) = \mathrm{Ang}(\bar{q}, q, r)\,\big(z_q z_{\bar{q}}^{1/2}\langle qr\rangle\big) \tag{111}$$
$$\mathcal{S}_g(\bar{q}^-, q^+, g_r^+) = \mathrm{Ang}(\bar{q}, q, r)\,\big(z_q^{1/2} z_{\bar{q}}\langle r\bar{q}\rangle\big) \tag{112}$$
$$\mathcal{S}_g(\bar{q}^-, q^+, g_r^-) = \mathrm{Sqr}(\bar{q}, q, r)\,\big(-z_q^{3/2}[r\bar{q}]\big) \tag{113}$$
$$\mathcal{S}_g(g_r^+, \bar{q}^-, q^+) = \mathrm{Ang}(\bar{q}, q, r)\,\big(z_{\bar{q}}^{3/2}\langle qr\rangle\big) \tag{114}$$
$$\mathcal{S}_g(g_r^-, \bar{q}^-, q^+) = \mathrm{Sqr}(\bar{q}, q, r)\,\big(-z_q z_{\bar{q}}^{1/2}[qr]\big) \tag{115}$$

For the squared splitting function summed over all colors and the final-state helicities, we find

$$\sum_{h_q,h_{\bar{q}}}\sum_{a_r,j,i}\big|\Lambda^{-1}\mathcal{A}_{ji}^{a_1a_2\cdots a_na_r}\big|^2 \to \mathrm{Tr}(\Omega\Omega^\dagger)$$
$$\times\,(z_q^2+z_{\bar{q}}^2)\frac{|\bar{q}_\perp+q_\perp+r_\perp|^2}{|s_{qr}||s_{q\bar{q}}||s_{r\bar{q}}|}\Big\{C_F\big[z_q|s_{r\bar{q}}|+z_{\bar{q}}|s_{rq}|\big]-\frac{1}{2N_c}\big[s_{q\bar{q}}+z_q s_{r\bar{q}}+z_{\bar{q}}s_{rq}\big]\Big\}\,. \tag{116}$$

## 3.7 General derivation for auxiliary quark-antiquark pair

We need to study amplitudes for which a gluon besides the quark-antiquark pair gets a divergent momentum component for $P^\mu$.

### 3.7.1 A gluon with divergent momentum

We do so by considering planar multi-gluon off-shell currents with such a gluon. An $(j-i+1)$-gluon off-shell current $\mathcal{J}^\mu_{i,j}$ is an $(j-i+2)$-gluon Green function with $(j-i+1)$ on-shell, amputated, legs. The subscript indicates that it includes the on-shell gluons $i$ to $j$. The one off-shell leg includes a propagator, and we will denote the off-shell current without this propagator with a tilde:

$$\mathcal{J}^\mu_{i,j} = \frac{-iD^\mu_\nu(K_{i,j},\ell)}{K^2_{i,j}}\,\tilde{\mathcal{J}}^\nu_{i,j} \quad , \quad K^\mu_{i,j} = \sum_{l=i}^{j} K^\mu_l \quad . \tag{117}$$

The symbol $D^{\mu\nu}$ refers to the propagator numerator, given by

$$D^{\mu\nu} = g^{\mu\nu} \quad \text{in the Feynman gauge} \tag{118}$$

or

$$D^{\mu\nu}(K,\ell) = g^{\mu\nu} - \frac{K^\mu\ell^\nu+\ell^\mu K^\nu}{\ell\cdot K} + \ell^2\frac{K^\mu K^\nu}{(\ell\cdot K)^2} \quad \text{in the axial gauge with gauge vector } \ell^\mu. \tag{119}$$

Tilde-currents satisfy current conservation

$$K_{i,j}\cdot\tilde{\mathcal{J}}_{i,j} = 0\,. \tag{120}$$

The currents with propagator satisfy this only in the Feynman gauge, but do satisfy

$$\ell\cdot\mathcal{J}_{i,j} = 0 \quad \text{in the axial gauge if} \quad \ell\cdot K_{i,j} \neq 0\,. \tag{121}$$

A typical situation in which $\ell\cdot K = 0$ is when the momentum has the form

$$K^\mu = z\ell^\mu + K_\perp^\mu \quad \text{with} \quad \ell\cdot\ell = \ell\cdot K_\perp = 0\,. \tag{122}$$

Had we chosen another gauge vector, $\ell'^\mu$, also satisfying $\ell'\cdot K_\perp = 0$ but with $\ell'\cdot\ell \neq 0$, then

$$\ell_\mu D^{\mu\nu}(K,\ell') = \ell^\nu - \frac{(\ell\cdot\ell')K^\nu}{z(\ell'\cdot\ell)} = -\frac{1}{z}\,K_\perp^\nu\,. \tag{123}$$

Now we can imagine that

$$\ell'^\mu = \ell^\mu + \epsilon\bar{\ell}^\mu \quad \text{with} \quad \bar{\ell}\cdot\bar{\ell} = \bar{\ell}\cdot K_\perp = 0 \quad \text{but} \quad \bar{\ell}\cdot\ell \neq 0 \quad , \tag{124}$$

and see that Eq. (123) holds for $\epsilon \to 0$. We conclude that

$$\ell\cdot\mathcal{J}_{i,j} = \begin{cases} 0 & \text{if} \quad \ell\cdot K_{i,j} \neq 0 \\ i\dfrac{K_\perp\cdot\tilde{\mathcal{J}}_{i,j}}{zK_\perp^2} & \text{if} \quad K_{i,j}^\mu = z\ell^\mu + K_\perp^\mu \end{cases} \quad \text{in the axial gauge with gauge vector } \ell^\mu. \tag{125}$$

We will need to take $\epsilon$ a bit more serious, and also encounter the situation with

$$K^\mu = z\ell^\mu + K_\perp^\mu + \epsilon^2\bar{\ell}^\mu\,, \tag{126}$$

so $\ell\cdot K = \mathcal{O}(\epsilon^2)$, and

$$\ell_\mu D^{\mu\nu}(K,\ell') = \ell^\nu - \frac{\epsilon K^\nu + \epsilon^2\ell'^\nu}{z\epsilon + \epsilon^2} + 2\epsilon\frac{\epsilon^2 K^\nu}{(z\epsilon+\epsilon^2)^2} = -\frac{1}{z}\,K_\perp^\nu + \mathcal{O}(\epsilon)\,, \tag{127}$$

while for other momenta with $\ell\cdot K = \mathcal{O}(\epsilon^0)$:

$$\ell_\mu D^{\mu\nu}(K,\ell') = \mathcal{O}(\epsilon)\,. \tag{128}$$

A one-point current by definition is the polarization vector of the external gluon it represent, and does not include a propagator, and it is understood for the equal-index currents that

$$\mathcal{J}_{i,i}^\mu = \tilde{\mathcal{J}}_{i,i}^\mu = \mathcal{J}_i^\mu = \tilde{\mathcal{J}}_i^\mu = \varepsilon^\mu(K_i)\,. \tag{129}$$

Polarizations may be chosen to satisfy $q\cdot\varepsilon$, with an auxiliary momentum $q^\mu$ different from $\ell^\mu$, but in practice it is most often best to choose the gauge vector from the propagator.

If an off-shell current includes a set of external gluons the momenta of which add up to a light-like momentum, so $K_{1,n}^2 = 0$, then the contraction of the tilde-current with a polarization vector with the opposite momentum, $\varepsilon_\mu(-K_{1,n})\tilde{\mathcal{J}}_{1,n}^\mu$, results in an $(n+1)$-gluon amplitude. Off-shell currents satisfy the well-know planar Berends-Giele recursion [3] of Fig. 1. This can be

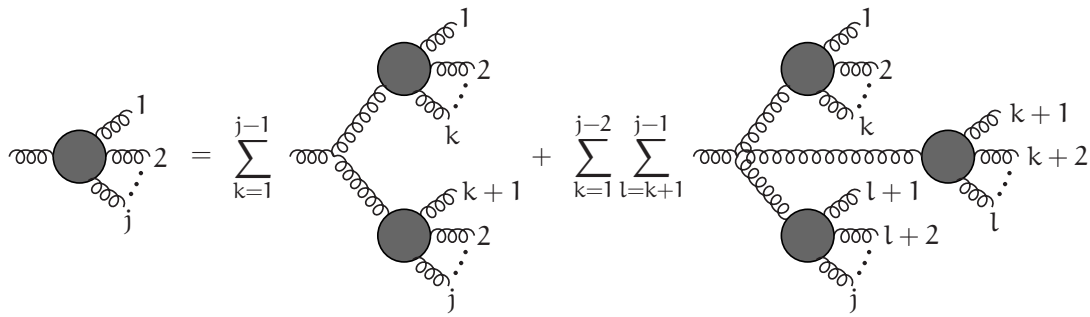



Figure 1: Berends-Giele recursive relation for gluon currents. Enumerated partons are on-shell.

considered a relation for the tilde-currents if the explicit internal lines in the figure are interpreted as propagators, as usual in Feynman graphs, while the external ones, including the off-shell one, are not. Typically, however, currents on the right-hand-side are assumed to include the propagators.

The vertices follow the planar Feynman rules, and writing the three-point vertex terms explicitly (and ignoring the coupling constant), we have

$$\begin{aligned}\tilde{\mathcal{J}}^{\mu}_{1,j} = \frac{i}{\sqrt{2}} \sum_{k=1}^{j-1} \Big[ & (K_{1,k} - K_{k+1,j})^{\mu} (\mathcal{J}_{1,k} \cdot \mathcal{J}_{k+1,j}) \\ & + \big((2K_{k+1,j} - K_{1,k}) \cdot \mathcal{J}_{1,k}\big) \mathcal{J}^{\mu}_{k+1,j} - \big((2K_{1,k} - K_{k+1,j}) \cdot \mathcal{J}_{k+1,j}\big) \mathcal{J}^{\mu}_{1,k} \Big] \\ + \big(\text{four-point vertex terms}\big) \, . \end{aligned} \tag{130}$$

Now we consider the situation in which gluon number 1 has momentum

$$K_1^{\mu} = \Lambda P^{\mu} + R_1^{\mu} \quad \text{with} \quad K_1^2 = 0 \quad \Rightarrow \quad P \cdot R_1 = \frac{R_1^2}{2\Lambda} = \mathcal{O}\big(\Lambda^{-1}\big) \, , \tag{131}$$

where $\Lambda$ becomes large. The polarization vector of this external gluon is of course still $\mathcal{O}\big(\Lambda^0\big)$. We will show now by induction that all currents involving this gluon are still $\mathcal{O}\big(\Lambda^0\big)$. For currents including gluon number 1, the relation becomes at leading order in $\Lambda$

$$\tilde{\mathcal{J}}^{\mu}_{1,j} = \frac{i}{\sqrt{2}} \sum_{k=1}^{j-1} \Big[ \Lambda P^{\mu} (\mathcal{J}_{1,k} \cdot \mathcal{J}_{k+1,j}) - \Lambda \big(P \cdot \mathcal{J}_{1,k}\big) \mathcal{J}^{\mu}_{k+1,j} - 2\Lambda \big(P \cdot \mathcal{J}_{k+1,j}\big) \mathcal{J}^{\mu}_{1,k} \Big] \; + \; \cdots \, . \tag{132}$$

Realize that the four-point vertices do not involve momenta and cannot produce terms of leading order in $\Lambda$. Possible gauge terms in the propagator to construct $\mathcal{J}^{\mu}_{1,j}$ from Eq. (117) are $\mathcal{O}\big(\Lambda^0\big)$, while the kinematical factor introduces $K^2_{1,j} = 2\Lambda P \cdot (R_1 + K_{2,j}) + \mathcal{O}\big(\Lambda^0\big)$ in the denominator. Thus the current $\mathcal{J}^{\mu}_{1,j}$ is of $\mathcal{O}\big(\Lambda^0\big)$.

Now, we choose the axial gauge with gauge vector

$$\ell'^{\mu} = P^{\mu} + \Lambda^{-\frac{1}{2}} \, \bar{P}^{\mu} \, , \tag{133}$$

so compared to the previous section we have $\epsilon = \Lambda^{-\frac{1}{2}}$ and $P\cdot\mathcal{J}_{a,b} = \mathcal{O}\big(\Lambda^{-\frac{1}{2}}\big)$. The recursion reduces to

$$\tilde{\mathcal{J}}^{\mu}_{1,j} = \frac{i\Lambda P^{\mu}}{\sqrt{2}}\sum_{k=1}^{j-1}(\mathcal{J}_{1,k}\cdot\mathcal{J}_{k+1,j}) \;-\; i\Lambda\sqrt{2}(P\cdot\mathcal{J}_{2,j})\mathcal{J}^{\mu}_{1}\,\theta_{j=n} \;+\; \mathcal{O}\big(\Lambda^{\frac{1}{2}}\big)\;. \tag{134}$$

We kept the term with $P\cdot\mathcal{J}_{2,j}$ for $j = n$ because, as we will see, that one is not subleading due to $P\cdot K_{2,n} = \mathcal{O}\big(\Lambda^{-1}\big)$. We are allowed now to explicitly write $\mathcal{O}\big(\Lambda^{\frac{1}{2}}\big)$ instead of ellipses like in Eq. (132), because we just proved that $\mathcal{J}^{\mu}_{1,j} = \mathcal{O}\big(\Lambda^{0}\big)$. It may be questionable to choose $P^{\mu}$ as the auxiliary vector for the polarization vector of gluon number 1, however we do have

$$K_1\cdot\mathcal{J}_1 = 0 \quad\Rightarrow\quad \Lambda P\cdot\mathcal{J}_1 = -R_1\cdot\mathcal{J}_1 = \mathcal{O}\big(\Lambda^{0}\big)\;, \tag{135}$$

so to leading order in $\Lambda$, the reduced recursion relation also holds for the term with $k = 1$ when we choose another auxiliary vector for $\mathcal{J}^{\mu}_{1}$. Thus we see that

$$\tilde{\mathcal{J}}^{\mu}_{1,k} = \Lambda C_{1,k}P^{\mu} + \mathcal{O}\big(\Lambda^{\frac{1}{2}}\big) \quad\text{and}\quad \mathcal{J}^{\mu}_{1,k} = \frac{C_{1,k}}{2P\cdot K_{1,k}}\,P^{\mu} + \mathcal{O}\big(\Lambda^{-\frac{1}{2}}\big) \quad\text{for}\quad 1 < k < n \quad , \tag{136}$$

and for some coefficients $C_{1,k}$ of $\mathcal{O}\big(\Lambda^{0}\big)$. Inserting Eq. (136) into Eq. (134) and using that $P\cdot\mathcal{J}_{k+1,j} = \mathcal{O}\big(\Lambda^{-\frac{1}{2}}\big)$ for $k > 1$, we find

$$\tilde{\mathcal{J}}^{\mu}_{1,j} = \frac{i\Lambda P^{\mu}}{\sqrt{2}}(\mathcal{J}_1\cdot\mathcal{J}_{2,j}) - i\Lambda\sqrt{2}(P\cdot\mathcal{J}_{2,j})\mathcal{J}^{\mu}_{1}\,\theta_{j=n} + \mathcal{O}\big(\Lambda^{\frac{1}{2}}\big)\;. \tag{137}$$

Now let the total number of gluons be $n+1$, and let both gluon number 1 and $n+1$ have divergent momentum components

$$K^{\mu}_{1} = (x-\Lambda)P^{\mu} + \beta k^{\mu}_{\perp} - \frac{\beta^2 k^2_{\perp}}{s(x-\Lambda)}\bar{P}^{\mu} \tag{138}$$

$$K^{\mu}_{n+1} = \Lambda P^{\mu} + (1-\beta)k^{\mu}_{\perp} - \frac{(1-\beta)^2 k^2_{\perp}}{s\Lambda}\bar{P}^{\mu}\;. \tag{139}$$

Momentum conservation states

$$K^{\mu}_{2,n} = -K^{\mu}_{1} - K^{\mu}_{n+1} = -xP^{\mu} - k^{\mu}_{\perp} + \left[\frac{\Lambda}{x-\Lambda}\beta^2 + (1-\beta)^2\right]\frac{k^2_{\perp}}{s\Lambda}\,\bar{P}^{\mu}\;. \tag{140}$$

We could choose $\beta$ such that the $\bar{P}$ component vanishes and avoid the need for the $\bar{P}$ component in the gauge vector Eq. (133), but this will not be possible anymore later on, so will not do it now either. Consequently, we have $K^2_{2,n} = k^2_{\perp} + \mathcal{O}\big(\Lambda^{-1}\big)$ and $P\cdot K_{2,n} = \mathcal{O}\big(\Lambda^{-1}\big)$, so indeed $P\cdot\mathcal{J}_{2,n}$ is not subleading:

$$P\cdot\mathcal{J}_{2,n} = i\frac{k_{\perp}\cdot\tilde{\mathcal{J}}_{2,n}}{xk^2_{\perp}} + \mathcal{O}\big(\Lambda^{-\frac{1}{2}}\big)\;. \tag{141}$$

Given the momentum $K^{\mu}_{n+1}$, it is easy to see, like in (135), that the polarization vector for this gluon satisfies $P\cdot\mathcal{J}_{n+1} = \mathcal{O}\big(\Lambda^{-1}\big)$, so

$$\mathcal{J}_{n+1}\cdot\tilde{\mathcal{J}}_{1,n} = -i\Lambda\sqrt{2}(P\cdot\mathcal{J}_{2,n})(\mathcal{J}_{n+1}\cdot\mathcal{J}_1) + \mathcal{O}\big(\Lambda^{\frac{1}{2}}\big)\;. \tag{142}$$

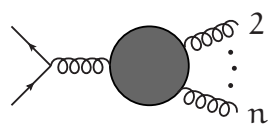
Choosing opposite helicities for $\mathcal{J}_1^\mu$ and $\mathcal{J}_{n+1}^\mu$, we have

$$\mathcal{J}_{n+1}\cdot\mathcal{J}_1 = -1 + \mathcal{O}\big(\Lambda^{-1}\big)\ , \tag{143}$$

and

$$\mathcal{J}_{n+1}\cdot\tilde{\mathcal{J}}_{1,n} = i\Lambda\sqrt{2}(P\cdot\mathcal{J}_{2,n}) + \mathcal{O}\big(\Lambda^{\frac{1}{2}}\big) = -\Lambda\frac{\sqrt{2}}{xk_\perp^2}(k_\perp\cdot\tilde{\mathcal{J}}_{2,n}) + \mathcal{O}\big(\Lambda^{\frac{1}{2}}\big)\ , \tag{144}$$

which is exactly what we expect for the amplitude with one space-like gluon with momentum $k^\mu = xP^\mu + k_\perp^\mu$ and on-shell gluons 2 to $n$. We essentially derived, at tree-level, that this amplitude can be obtained with auxiliary gluons instead of an auxiliary quark-antiquark pair. This can be understood from the well-known fact that, in case an auxiliary quark-antiquark pair is used, the auxiliary quark line inside Feynman graphs to leading order in $\Lambda$ becomes an eikonal line with quark-gluon vertex proportional to $P^\mu$, and that in the axial gauge with gauge vector $P^\mu$ the only graphs that contribute are of the form




. (145)

That is the amplitude is given by off-shell current containing all gluons attached to the quark-antiquark pair with a single vertex, leading to Eq. (144).

### 3.7.2 Gluon and quark-antiquark pair with divergent $P$ coefficient

We consider the process $\emptyset \to \bar{q}qg_{r=1}g_2\ldots g_n$, where the first three parton momenta have divergent $P$ coefficients:

$$K_{\bar{q}}^\mu = -(x+\Lambda)P^\mu\ \ ,\ \ K_q^\mu = z_q\Lambda P^\mu + q_\perp^\mu - \frac{q_\perp^2}{sz_q\Lambda}\bar{P}^\mu\ \ ,\ \ K_r^\mu = z_r\Lambda P^\mu + r_\perp^\mu - \frac{r_\perp^2}{sz_r\Lambda}\bar{P}^\mu\ , \tag{146}$$

with

$$z_r + z_q = 1\ . \tag{147}$$

So we have, to $\mathcal{O}\big(\Lambda^{-1}\big)$,

$$K_{\bar{q}}^\mu + K_q^\mu + K_r^\mu = -xP^\mu + q_\perp^\mu + r_\perp^\mu \quad , \quad (K_{\bar{q}} + K_q + K_r)^2 = (q_\perp + r_\perp)^2 \tag{148}$$

and

$$(K_{\bar{q}} + K_q)^2 = \frac{q_\perp^2}{z_q}\ \ ,\ \ (K_{\bar{q}} + K_r)^2 = \frac{r_\perp^2}{z_r}\ \ ,\ \ (K_q + K_r)^2 = (q_\perp + r_\perp)^2 - \frac{q_\perp^2}{z_q} - \frac{r_\perp^2}{z_r}\ . \tag{149}$$

It is important to observe that these three invariants do not depend on $\Lambda$ and do not produce a factor of $\Lambda$ in propagator denominators. The spinors of the momenta are give by

$$|\bar{q}\rangle = \sqrt{\Lambda + x}\,|P\rangle \qquad\qquad |\bar{q}] = -\sqrt{\Lambda + x}\,|P] \tag{150}$$

$$|q\rangle = \sqrt{z_q\Lambda}\,|P\rangle + \frac{\kappa_q^\star}{\sqrt{z_q\Lambda}\langle P\bar{P}\rangle}\,|\bar{P}\rangle \qquad |q] = \sqrt{z_q\Lambda}\,|P] + \frac{\kappa_q}{\sqrt{z_q\Lambda}[\bar{P}P]}\,|\bar{P}] \tag{151}$$

$$|r\rangle = \sqrt{z_r\Lambda}\,|P\rangle + \frac{\kappa_r^\star}{\sqrt{z_r\Lambda}\langle P\bar{P}\rangle}\,|\bar{P}\rangle \qquad |r] = \sqrt{z_r\Lambda}\,|P] + \frac{\kappa_r}{\sqrt{z_r\Lambda}[\bar{P}P]}\,|\bar{P}] \tag{152}$$

and the for polarization vectors for the radiative gluon we use $\bar{P}^\mu$ as auxiliary momentum, so

$$|r^-\rangle = |r\rangle \ , \ |r^-] = \frac{\sqrt{2}}{\sqrt{z_r}\Lambda[P\bar{P}]}\,|\bar{P}] \quad \text{and} \quad |r^+] = |r] \ , \ |r^+\rangle = \frac{\sqrt{2}}{\sqrt{z_r}\Lambda\langle\bar{P}P\rangle}\,|\bar{P}\rangle \ . \tag{153}$$

Spinor products

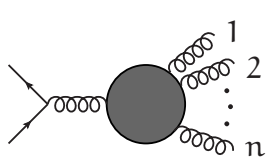
$$\begin{aligned}
\langle q\bar{q}\rangle &= -\frac{\kappa_q^\star}{\sqrt{z_q}} \quad , \quad [\bar{q}q] = \frac{\kappa_q}{\sqrt{z_q}} \\
\langle r\bar{q}\rangle &= -\frac{\kappa_r^\star}{\sqrt{z_r}} \quad , \quad [\bar{q}r] = \frac{\kappa_r}{\sqrt{z_r}} \\
\langle rq\rangle &= \sqrt{\frac{z_r}{z_q}}\,\kappa_q^\star - \sqrt{\frac{z_q}{z_r}}\,\kappa_r^\star \quad , \quad [qr] = \sqrt{\frac{z_r}{z_q}}\,\kappa_q - \sqrt{\frac{z_q}{z_r}}\,\kappa_r
\end{aligned} \tag{154}$$

$$\begin{aligned}
\langle r^-|r\rangle &= 0 \quad , \quad [r|r^-] = \sqrt{2} \\
\langle r^-|\bar{q}\rangle &= -\frac{\kappa_r^\star}{\sqrt{z_r}} \quad , \quad [\bar{q}|r^-] = -\sqrt{\frac{2}{z_r}} \\
\langle r^-|q\rangle &= \sqrt{\frac{z_r}{z_q}}\,\kappa_q^\star - \sqrt{\frac{z_q}{z_r}}\,\kappa_r^\star \quad , \quad [q|r^-] = \sqrt{\frac{2z_q}{z_r}}
\end{aligned} \tag{155}$$

$$\begin{aligned}
\langle r^+|r\rangle &= \sqrt{2} \quad , \quad [r|r^+] = 0 \\
\langle r^+|\bar{q}\rangle &= \sqrt{\frac{2}{z_r}} \quad , \quad [\bar{q}|r^+] = \frac{\kappa_r}{\sqrt{z_r}} \\
\langle r^+|q\rangle &= \sqrt{\frac{2z_q}{z_r}} \quad , \quad [q|r^+] = \sqrt{\frac{z_r}{z_q}}\,\kappa_q - \sqrt{\frac{z_q}{z_r}}\,\kappa_r
\end{aligned} \tag{156}$$

The amplitude for the process under consideration can be calculated by collecting contributions obtained by attaching multi-gluon off-shell currents $\mathcal{J}^\mu_{i,j}$ to the quark line. We saw in the previous section that all off-shell currents $\mathcal{J}^\mu_{1,j}$ are of $\mathcal{O}\big(\Lambda^0\big)$, because while all $\tilde{\mathcal{J}}^\mu_{1,j}$ (except $\tilde{\mathcal{J}}^\mu_1$) are of $\mathcal{O}\big(\Lambda^1\big)$, also all propagator denominators are: $K^2_{1,j} = 2\Lambda P\cdot K_{1,j} + \mathcal{O}\big(\Lambda^0\big)$. That is all except $\mathcal{J}^\mu_{1,n}$, because that one was not defined due to its momentum $K^\mu_{1,n} = -K^\mu_{n+1}$ being light-like. Now it is not light-like, but we just saw that the square of the momentum $K^\mu_{1,n} = -K^\mu_{\bar{q}} + K^\mu_q$ is still independent of $\Lambda$, and thus $\mathcal{J}^\mu_{1,n}$ is, like $\tilde{\mathcal{J}}^\mu_{1,n}$, of $\mathcal{O}\big(\Lambda^1\big)$. We conclude that contributions involving off-shell currents $\mathcal{J}^\mu_{1,j}$ other than $\mathcal{J}^\mu_{1,n}$ attached to the quark line are likely sub-leading. The exceptions are contributions involving the polarization vector $\mathcal{J}^\mu_1$ itself attached directly to the quark line, exactly because $(K_{\bar{q}} + K_1)^2$ and $(K_q + K_1)^2$ are independent of $\Lambda$.

The contribution with $\mathcal{J}^\mu_{1,n}$ looks as follows:

$\mathcal{C}(\bar{q}, q, 1) =$ 

(157)

and in the case $\bar{q}^+, q^-, g_1^-$ is given by

$$\mathcal{C}(\bar{q}^+, q^-, 1) = \frac{i}{\sqrt{2}} \langle q|\gamma_\mu \mathcal{J}^\mu_{1,n}|\bar{q}] = \frac{\langle q|\gamma^\mu|\bar{q}] D_{\mu\nu}\big(K_q + K_{\bar{q}}, P + \Lambda^{-\frac{1}{2}}\bar{P}\big)}{\sqrt{2}(K_{\bar{q}} + K_q)^2} \tilde{\mathcal{J}}^\nu_{1,n} \,. \tag{158}$$

Now we have $P\cdot(K_q + K_{\bar{q}}) = \mathcal{O}\big(\Lambda^{-1}\big)$, so we have to take care with the gauge projector. For the current we use the expression of Eq. (137) with $j = n$, but with $z_r\Lambda$ instead of just $\Lambda$. To $\mathcal{O}\big(\Lambda^{1/2}\big)$, we find

$$\mathcal{C}(\bar{q}^+, q^-, 1) = \frac{\langle q|\gamma_\mu|\bar{q}] - 2\frac{\sqrt{z_q}}{z_r}(K_q + K_{\bar{q}})_\mu}{\sqrt{2}(K_{\bar{q}} + K_q)^2} \frac{i z_r \Lambda}{\sqrt{2}} \big[(\mathcal{J}_1\cdot\mathcal{J}_{2,n})P^\mu - 2(P\cdot\mathcal{J}_{2,n})\mathcal{J}_1^\mu\big] \,. \tag{159}$$

Now, $\langle q|\not{P}|\bar{q}] = 0$ and $P\cdot(K_q + K_{\bar{q}}) = \mathcal{O}\big(\Lambda^{-1}\big)$, so

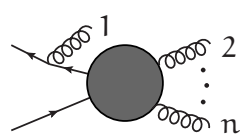
$$\mathcal{C}(\bar{q}^+, q^-, 1) = \frac{2\sqrt{z_q}(K_q + K_{\bar{q}})\cdot\mathcal{J}_1 - z_r\langle q|\gamma_\mu \mathcal{J}_1^\mu|\bar{q}]}{\sqrt{2}(K_{\bar{q}} + K_q)^2} i\sqrt{2}\Lambda(P\cdot\mathcal{J}_{2,n}) \,, \tag{160}$$

where we recognize $i\sqrt{2}\Lambda(P\cdot\mathcal{J}_{2,n})$ as $\Lambda$ times the $n$-gluon amplitude $\tilde{A}_{1,n}$ with one of them space-like with momentum $-xP^\mu + q_\perp^\mu + r_\perp^\mu$. Putting an explicit gluon helicity, we find

$$\begin{aligned}
\mathcal{C}(\bar{q}^+, q^-, 1^-) &= \frac{\sqrt{z_q}\langle q|1^-\rangle[1^-|q] + \sqrt{z_q}\langle \bar{q}|1^-\rangle[1^-|\bar{q}] - z_r\langle q|1^-\rangle[1^-|\bar{q}]}{\sqrt{2}(K_{\bar{q}} + K_q)^2} \Lambda\tilde{A}_{1,n} \\
&= \frac{\sqrt{z_q}\langle 1|q\rangle\sqrt{\frac{2z_q}{z_r}} + \sqrt{z_q}\frac{\sqrt{2}\kappa_r^\star}{z_r} + z_r\langle 1|q\rangle\sqrt{\frac{2}{z_r}}}{\sqrt{2}\langle \bar{q}q\rangle[q\bar{q}]} \Lambda\tilde{A}_{1,n} \\
&= \frac{\langle 1|q\rangle + \sqrt{\frac{z_q}{z_r}}\,\kappa_r^\star}{\sqrt{z_r}\langle \bar{q}q\rangle[q\bar{q}]} \Lambda\tilde{A}_{1,n} = \frac{\sqrt{\frac{z_r}{z_q}}\,\kappa_q^\star}{\sqrt{z_r}\langle \bar{q}q\rangle[q\bar{q}]} \Lambda\tilde{A}_{1,n} \,.
\end{aligned} \tag{161}$$

The other leading contribution for the case $\bar{q}^+, q^-, g_1^-$ looks like

$$\mathcal{D}(\bar{q}, q, 1) = \tag{162}$$




where, if $z_r$ were zero, $\sqrt{\Lambda}\langle P||\mathcal{J}_{2,n}\rangle$ would give $\Lambda$ times the amplitude for $n$ gluons, with one of them space-like with momentum $-xP^\mu + q_\perp^\mu$. Now, we have

$$\begin{aligned}
\mathcal{D}(\bar{q}^+, q^-, 1^-) &= \frac{i}{\sqrt{2}}\langle q|1^-\rangle[1^-|\, i\, \frac{|q]\langle q| + |1]\langle 1|}{\langle q1\rangle[1q]}\, |\mathcal{J}_{2,n}\rangle \\
&= \frac{-1}{\sqrt{2}[1q]}\left(\frac{-\sqrt{2z_q}}{\sqrt{z_r}}\langle q|\mathcal{J}_{2,n}\rangle - \sqrt{2}\langle 1|\mathcal{J}_{2,n}\rangle\right) \\
&= \frac{1}{[1q]}\left(\sqrt{z_q\Lambda}\sqrt{\frac{z_q}{z_r}} + \sqrt{z_r\Lambda}\right)\langle P|\mathcal{J}_{2,n}\rangle = \frac{1}{\sqrt{z_r}\,[1q]}\Lambda\tilde{A}_{1,n} \,.
\end{aligned} \tag{163}$$

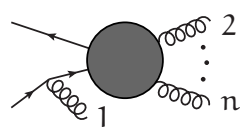
With the positive-helicity gluon, we get

$$\begin{aligned}
\mathcal{C}(\bar{q}^+, q^-, 1^+) &= \frac{\sqrt{z_q}\langle q|1^+\rangle[1^+|q] + \sqrt{z_q}\langle \bar{q}|1^+\rangle[1^+|\bar{q}] - z_r\langle q|1^+\rangle[1^+|\bar{q}]}{\sqrt{2}(K_{\bar{q}} + K_q)^2}\,\Lambda\tilde{A}_{1,n} \\
&= \frac{\sqrt{z_q}(-\sqrt{\frac{2z_q}{z_r}})[1q] + \sqrt{z_q}(-\sqrt{\frac{2}{z_r}})(-\frac{\kappa_r}{\sqrt{z_r}}) - z_r(-\sqrt{\frac{2z_q}{z_r}})(-\frac{\kappa_r}{\sqrt{z_r}})}{\sqrt{2}(K_{\bar{q}} + K_q)^2}\,\Lambda\tilde{A}_{1,n} \\
&= z_q\,\frac{\sqrt{\frac{z_r}{z_q}}\kappa_q}{\sqrt{z_r}\langle \bar{q}q\rangle[q\bar{q}]}\,\Lambda\tilde{A}_{1,n}\,. \qquad (164)
\end{aligned}$$

and

$$\begin{aligned}
\mathcal{D}(\bar{q}^+, q^-, 1^+) &= \frac{i}{\sqrt{2}}\langle q||1^+\rangle[1^+|\,i\,\frac{|q]\langle q| + |1]\langle 1|}{(K_q + K_1)^2}\,|\mathcal{J}_{2,n}\rangle \\
&= \frac{-1}{\sqrt{2}(K_q + K_1)^2}\left(-\sqrt{\frac{2z_q}{z_r}}\right)([1q]\langle q||\mathcal{J}_{2,n}\rangle) \\
&= \frac{1}{\langle q1\rangle[1q]}\left(\sqrt{\frac{z_q}{z_r}}\right)\left([1q]\sqrt{z_q\Lambda}\langle P||\mathcal{J}_{2,n}\rangle\right) \\
&= \frac{z_q}{\sqrt{z_r}\langle q1\rangle}\,\Lambda\tilde{A}_{1,n}\,. \qquad (165)
\end{aligned}$$

Let us also consider the situation when gluon 1 is attached to the external anti-quark. This contributes to the partial amplitude for another permutation of the gluons:

$\bar{\mathcal{D}}(\bar{q}, q, 1) =$ 

(166)

For the minus-helicity we get

$$\begin{aligned}
\bar{\mathcal{D}}(\bar{q}^+, q^-, 1^-) &= \frac{-1}{\sqrt{2}}[\mathcal{J}_{2,n}|\frac{|1]\langle 1| + |\bar{q}]\langle \bar{q}|}{\langle 1\bar{q}\rangle[\bar{q}1]}|1^-\rangle[1^-|\bar{q}] = \frac{1}{\sqrt{2}}[\mathcal{J}_{2,n}|\bar{q}]\frac{[1^-|\bar{q}]}{[\bar{q}1]} \\
&= -\frac{\sqrt{\Lambda}}{\sqrt{2}}[\mathcal{J}_{2,n}|P]\sqrt{\frac{2}{z_r}}\frac{1}{[\bar{q}1]} = \frac{\sqrt{z_q}}{\sqrt{z_r}\,[1\bar{q}]}\,\Lambda\tilde{A}_{1,n} \qquad (167)
\end{aligned}$$

where the factor $\sqrt{z_q}$ comes from the fact that $[\mathcal{J}_{2,n}|$ is terminated from the left by $\langle q|$ rather than $\sqrt{\Lambda}\langle P|$. For the plus-helicity we get

$$\begin{aligned}
\bar{\mathcal{D}}(\bar{q}^+, q^-, 1^+) &= \frac{-1}{\sqrt{2}}[\mathcal{J}_{2,n}|\frac{|1]\langle 1| + |\bar{q}]\langle \bar{q}|}{\langle 1\bar{q}\rangle[\bar{q}1]}|1^+\rangle[1^+|\bar{q}] \\
&= \frac{1}{\sqrt{2}}\frac{1}{\langle 1\bar{q}\rangle}\left(-\sqrt{z_r\Lambda}[\mathcal{J}_{2,n}|P]\sqrt{2} + \sqrt{\Lambda}[\mathcal{J}_{2,n}|P]\sqrt{\frac{2}{z_r}}\right) \\
&= \frac{\sqrt{z_q}}{\langle 1\bar{q}\rangle}\left(-\sqrt{z_r} + \frac{1}{\sqrt{z_r}}\right)\Lambda\tilde{A}_{1,n} = \frac{z_q^{3/2}}{\sqrt{z_r}\langle 1\bar{q}\rangle}\,\Lambda\tilde{A}_{1,n} \qquad (168)
\end{aligned}$$

Using these, we find

$$\mathcal{S}(\bar{q}^+, q^-, 1^-) \equiv \frac{\mathcal{D}(\bar{q}^+, q^-, 1^-) - \mathcal{C}(\bar{q}^+, q^-, 1^-)}{\Lambda \tilde{A}_{1,n}} = -z_q^{1/2} \frac{\kappa_q + \kappa_r}{\sqrt{z_r}[\bar{q}q][q1]} \tag{169}$$

$$\mathcal{S}(\bar{q}^+, q^-, 1^+) \equiv \frac{\mathcal{D}(\bar{q}^+, q^-, 1^+) - \mathcal{C}(\bar{q}^+, q^-, 1^+)}{\Lambda \tilde{A}_{1,n}} = z_q^{3/2} \frac{\kappa_q^\star + \kappa_r^\star}{\sqrt{z_r}\langle \bar{q}q\rangle\langle q1\rangle} \tag{170}$$

$$\bar{\mathcal{S}}(\bar{q}^+, q^-, 1^-) \equiv \frac{\bar{\mathcal{D}}(\bar{q}^+, q^-, 1^-) + \mathcal{C}(\bar{q}^+, q^-, 1^-)}{\Lambda \tilde{A}_{1,n}} = \frac{\kappa_r + \kappa_q}{\sqrt{z_r}[q\bar{q}][\bar{q}1]} \tag{171}$$

$$\bar{\mathcal{S}}(\bar{q}^+, q^-, 1^+) \equiv \frac{\bar{\mathcal{D}}(\bar{q}^+, q^-, 1^+) + \mathcal{C}(\bar{q}^+, q^-, 1^+)}{\Lambda \tilde{A}_{1,n}} = z_q \frac{\kappa_r^\star + \kappa_q^\star}{\sqrt{z_r}\langle \bar{q}q\rangle\langle 1\bar{q}\rangle} \,. \tag{172}$$

# 4 Real radiation contribution for $x_r < x_{\text{thrs}}$

Now we consider the real-radiation contribution for $x_r < x_{\text{thrs}}$:

$$\begin{aligned}
I_R^< &= \int dx_k\, d^4q\delta(q^2)\, d^4r\delta(r^2)\, \theta(x_{\text{thrs}} - x_r)\, \delta(x_k + x_q + x_r - \Lambda) \\
&\quad \times d\Phi_{1,n}\, \delta^4(\Lambda P + p_{n+1} - q - r - p_{1,n})\, \mathcal{M}\big(\Lambda P, q, r, \{p_i\}\big) \\
&= \int dx_k\, \frac{dx_q}{2x_q} d\bar{x}_q d^2q_\perp \delta\left(\bar{x}_q - \frac{|q_\perp|^2}{sx_q}\right) \frac{dx_r}{2x_r} d\bar{x}_r d^2r_\perp \delta\left(\bar{x}_r - \frac{|r_\perp|^2}{sx_r}\right) \\
&\quad \times \theta(x_{\text{thrs}} - x_r)\, \delta(x_k + x_q + x_r - \Lambda) \\
&\quad \times d\Phi_{1,n}\, \delta^4(x_k P - q_\perp - r_\perp - (\bar{x}_q + \bar{x}_r)\bar{P} + p_{n+1} - p_{1,n})\, \mathcal{M}\big(\Lambda P, q, r, \{p_i\}\big) \,.
\end{aligned} \tag{173}$$

Performing the $x_q$-integral eliminating the $\delta(x_k + x_q + x_r - \Lambda)$, we get

$$\begin{aligned}
I_R^< &= \frac{1}{4} \int dx_k\, d\bar{x}_q d^2q_\perp \delta\left(\bar{x}_q - \frac{|q_\perp|^2}{s(\Lambda - x_k - x_r)}\right) \frac{dx_r}{x_r} d\bar{x}_r d^2r_\perp \delta\left(\bar{x}_r - \frac{|r_\perp|^2}{sx_r}\right) \theta(x_{\text{thrs}} - x_r) \\
&\quad \times d\Phi_{1,n}\, \delta^4(x_k P - q_\perp - r_\perp - (\bar{x}_q + \bar{x}_r)\bar{P} + p_{n+1} - p_{1,n}) \\
&\quad \times \frac{\mathcal{M}\big(\Lambda P,\, (\Lambda - x_k - x_r)P + \bar{x}_q \bar{P} + q_\perp, r, \{p_i\}\big)}{\Lambda - x_k - x_r} \,.
\end{aligned} \tag{174}$$

Now we take $\Lambda \to \infty$, implying $\bar{x}_q \to 0$, and we find

$$\begin{aligned}
I_R^{<,\text{lim}} &= \frac{\Lambda}{4} \int dx_k\, d^2q_\perp\, \frac{dx_r}{x_r} d\bar{x}_r d^2r_\perp \delta\left(\bar{x}_r - \frac{|r_\perp|^2}{sx_r}\right) \theta(x_{\text{thrs}} - x_r) \\
&\quad \times d\Phi_{1,n}\, \delta^4(x_k P - q_\perp - r_\perp - \bar{x}_r \bar{P} + p_{n+1} - p_{1,n})\, \mathcal{M}^*\big((x_k + x_r)P - q_\perp, r, \{p_i\}\big) \,.
\end{aligned} \tag{175}$$

We know that $\mathcal{M}^*$ has a singularity at $|r_\perp| = 0$ of the form $1/P\cdot r = 2/(s\bar{x}_r) = 2x_r/|r_\perp|^2$, so we are going to take this limit in the matrix element and the phase space. We have

$$\mathcal{M}^*\big(xP - q_\perp, r, \{p_i\}\big) \xrightarrow{r \to yP} \mathcal{Q}(x, y, P, q_\perp, r)\, \mathcal{M}^*\big((x+y)P - q_\perp, \{p_i\}\big) \tag{176}$$

Besides the $1/P\cdot r$ singularity, the function $\mathcal{Q}$ also contains terms that stay finite only because $|q_\perp| > 0$, and give rise to logarithms of $|q_\perp|$. Realize that we have $x = x_k + x_r$ and $y = -x_r$ in our instance of the limit (176), so

$$I^{<,\lim}_{R,\text{sing}} = \frac{\Lambda}{4}\int dx_k\, d^2q_\perp\, \frac{dx_r}{x_r} d\bar{x}_r d^2r_\perp \delta\left(\bar{x}_r - \frac{|r_\perp|^2}{sx_r}\right)\, \theta(x_{\text{thrs}} - x_r)\, \mathcal{Q}(x_k + x_r, -x_r, P, q_\perp, r_\perp)$$
$$\times\, d\Phi_{1,n}\, \delta^4(x_k P - q_\perp + p_{n+1} - p_{1,n})\, \mathcal{M}^*\big(x_k P - q_\perp, \{p_i\}\big)\,. \tag{177}$$

We see that the phase space and matrix element completely factorize from the $r$-integration, even without a form of convolution with a "$z$ variable" like usually in collinear factorization. This will come back with the interpretation of the variable $x_k$. Thus we just need to calculate

$$\int \frac{dx_r}{x_r}\, \theta(x_{\text{thrs}} - x_r)\, d^2r_\perp\, \theta\big(sx_r - |r_\perp|^2\big)\, \mathcal{Q}(x_k + x_r, -x_r, P, q_\perp, r_\perp)\,. \tag{178}$$

We need to use dimensional regularization to deal with the singularity

$$I_1(x_r) = \int d^{2-2\varepsilon} r_\perp\, \theta\big(sx_r - |r_\perp|^2\big)\, \frac{1}{|r_\perp|^2} = \frac{2\pi^{1-\varepsilon}}{\Gamma(1-\varepsilon)}\, \frac{(sx_r)^{-\varepsilon}}{-2\varepsilon} \tag{179}$$

The integrals giving rise to logarithms of $|q_\perp|$ will be of the type

$$I_2(x_r) = \int_0^{2\pi} d\phi \int_0^{\sqrt{sx_r}} d|r_\perp||r_\perp|\, \frac{1}{|r_\perp|^2 - 2a|q_\perp||r_\perp|\cos\phi + a|q_\perp|^2} \tag{180}$$

$$I_3(x_r) = \int_0^{2\pi} d\phi \int_0^{\sqrt{sx_r}} d|r_\perp||r_\perp|\, \frac{|r_\perp|^2}{\big(|r_\perp|^2 - 2a|q_\perp||r_\perp|\cos\phi + a|q_\perp|^2\big)^2} \tag{181}$$

where $a = x_r/(x_k + x_r)$, so $a^2\cos\phi - a < 0$ and the integrals are finite. The logarithmic behavior with $|q_\perp|$ comes from the lower integration limit $|r_\perp| = 0$, and we find

$$I_2(x_r) \approx -\pi \log\left(\frac{|q_\perp|^2}{(x_k + x_r)s}\right) \quad , \quad I_2(x_r) \approx -\pi \log\left(\frac{|q_\perp|^2}{(x_k + x_r)s}\right) \tag{182}$$

## 4.1 Collinear limits

Starting from an MHV amplitude with a space-like gluon with momentum $k_1 = p_1 + k_T$ and which is singlular when $p_1$ and $p_2$ become collinear,

$$A_n\big(1^*, 2^+, 3^-, 4^+, \ldots, n^+\big) = \frac{i|k_T|}{\kappa^-}\frac{\langle p_1 p_3\rangle^4}{\langle p_1p_2\rangle\langle p_2p_3\rangle\langle p_3p_4\rangle\cdots\langle p_{n-1}p_n\rangle\langle p_np_1\rangle}\,, \tag{183}$$

we can easily extract the limit $p_1 \to xP$ and $p_2 \to yP$

$$A_n\big(1^*, 2^+, 3^-, 4^+, \ldots, n^+\big) \xrightarrow{p_1\to xP,\, p_2\to yP} \frac{1}{\langle p_1p_2\rangle}\frac{i|k_T|}{\kappa^-}\frac{x^2\langle Pp_3\rangle^4}{\sqrt{|y|}\langle Pp_3\rangle\langle p_3p_4\rangle\cdots\langle p_{n-1}p_n\rangle\sqrt{|x|}\langle p_nP\rangle}$$
$$= \frac{1}{\langle p_1p_2\rangle}\frac{x^2}{\sqrt{|xy|}|x+y|}\frac{i|k_T|}{\kappa^-}\frac{\langle (x+y)P|p_3\rangle^4}{\langle (x+y)P|p_3\rangle\langle p_3p_4\rangle\cdots\langle p_{n-1}p_n\rangle\langle p_n|(x+y)P\rangle}$$
$$= \frac{1}{\langle P|p_2\rangle}\frac{|x|}{\sqrt{|y|}|x+y|}A_{n-1}\big(1^*, 3^-, 4^+, \ldots, n^+\big) \quad \text{now with} \quad p_1 = (x+y)P\,. \tag{184}$$

For the opposite helicity there is $\sqrt{|y|}/y$ instead of $1/\sqrt{|y|}$, and also $1/[p_2P] = -1/[Pp_2]$. Assuming that

$$x > 0 \quad , \quad x + y > 0 \quad , \quad y < 0 \quad , \tag{185}$$

we get for the general case

$$A_n\big(g^*(k), g(r)^\pm, \dots\big)\big| \to \frac{1}{(Pr)_\mp} \frac{x}{\sqrt{|y|}(x+y)} A_{n-1}\big(g^*((x+y)P + k_T), \dots\big) \, . \tag{186}$$

where we denote

$$(Pr)_+ = [Pr] \quad , \quad (Pr)_- = \langle Pr \rangle \quad . \tag{187}$$

Terms that behave as $1/(k+r)^2$ or as $p{\cdot}r/(k+r)^4$ on the other hand are not singular, but only because $|k_T| > 0$. Thus they produce logs of $|k_T|$ in the integration of $r_T$. Terms in the amplitude with the denominator factor $s_{kr} = (k+r)^2$ are easily identified by considering the limit $s_{kr} \to 0$. The well-known factorization behavior of scattering amplitudes tells us that

$$A_n(g^*(k), g(r), \dots) \stackrel{s_{kr}\to 0}{\longrightarrow} \sum_{h=+,-} A_3\big(g^*(k), g(r), g(K)^h\big) \frac{i}{s_{kr}} A_{n-1}\big(g(-K)^{-h}, \dots\big) \tag{188}$$

where

$$K^\mu = -k^\mu - r^\mu \quad \text{is light-like} \, . \tag{189}$$

Since the equal-helicity amplitudes $A_3\big(g^*(k), g(r)^\pm, g(K)^\pm\big)$ are known to vanish, we can simplify the formula for explicit helicities of the radiative gluon

$$A_n(g^*(k), g(r)^\pm, \dots) \stackrel{s_{kr}\to 0}{\longrightarrow} \frac{i}{s_{kr}} A_3\big(g^*(k), g(r)^\pm, g(K)^\mp\big) A_{n-1}\big(g(-K)^\pm, \dots\big) \, . \tag{190}$$

The non-equal helicity amplitudes are given by [1]

$$A_3\big(g^*(k), g(r)^+, g(K)^-\big) = i\frac{|k_T|}{\kappa} \frac{[rp]^3}{[pK][Kr]} \, , \tag{191}$$

$$A_3\big(g^*(k), g(r)^-, g(K)^+\big) = -A_3\big(g^*(k), g(K)^+, g(r)^-\big) = -i\frac{|k_T|}{\kappa^\star} \frac{\langle rp\rangle^3}{\langle pK\rangle\langle Kr\rangle} \, . \tag{192}$$

Now we are interested in the limit $p^\mu \to xP^\mu$ and $r^\mu \to yP^\mu$ again. The two amplitudes above vanish in this limit, but we want to see how. Due to momentum conservation we also have $K^\mu \to -(x+y)P^\mu$ and equivalently $K^\mu \to -(x+y)/y r^\mu$, so

$$A_3\big(g^*(k), g(r)^-, g(K)^+\big) \to i\frac{|k_T|}{\kappa^\star} \frac{-|x|^{3/2}\langle rP\rangle^3}{\sqrt{|x|}\sqrt{\left|\frac{x+y}{y}\right|}\langle Pr\rangle\sqrt{|x+y|}\langle Pr\rangle} = i\frac{|k_T|}{\kappa^\star} \frac{\sqrt{|y|}\,|x|}{|x+y|} (Pr)_-$$

$$A_3\big(g^*(k), g(r)^+, g(K)^-\big) \to -i\frac{|k_T|}{\kappa^\star} \frac{y}{\sqrt{|y|}} \frac{x}{x+y} (Pr)_+ \tag{193}$$

Assuming Eq. (185) again, we find

$$A_n\big(g^*(k), g(r)^\pm, \dots\big)\big|_{\text{logs}} \to \frac{(Pr)_\pm}{s_{kr}} \frac{\sqrt{|y|}\,x}{x+y} \frac{|k_T|}{\kappa^\pm} A_{n-1}\big(g((x+y)P)^\pm, \dots\big)\,. \tag{194}$$

with

$$s_{kr} = (xP + k_T + r)^2\,. \tag{195}$$

So eventually, we find, using $(Pr)_\pm (Pr)_\mp = -s_{Pr}$,

$$A_n\big(g^*(k), g(r)^\pm, \dots\big) \to \frac{x}{\sqrt{|y|}(x+y)} \frac{1}{(Pr)_\mp} \Bigg[A_{n-1}\big(g^*((x+y)P + k_T), \dots\big) - |y| \frac{s_{Pr}}{s_{kr}} \frac{|k_T|}{\kappa^\pm} A_{n-1}\big(g((x+y)P)^\pm, \dots\big)\Bigg]\,. \tag{196}$$

Let us abbreviate this to

$$A^{*,\pm}_{n+1} \to \frac{x}{\sqrt{|y|}(x+y)} \frac{1}{(Pr)_\mp} \left[A^*_n - |y| \frac{s_{Pr}}{s_{kr}} \frac{|k_T|}{\kappa^\pm} A^\pm_n\right]\,. \tag{197}$$

For the square of the absolute value, we find, denoting complex conjugation with a $\star$,

$$\left|A^{*,\pm}_{n+1}\right|^2 \to \frac{x^2}{|y|(x+y)^2} \frac{1}{|s_{Pr}|} \Bigg\{\left|A^*_n\right|^2 + y^2 \frac{s^2_{Pr}}{s^2_{kr}} \left|A^\pm_n\right|^2 - |y| \frac{s_{Pr}}{s_{kr}} \left[A^*_n \left(\frac{|k_T|}{\kappa^\pm} A^\pm_n\right)^\star + \left(A^*_n\right)^\star \frac{|k_T|}{\kappa^\pm} A^\pm_n\right]\Bigg\} \tag{198}$$

For the terms producing logarithms in $|k_T|$, it is not essential for the amplitudes to be on-shell. The limit is formally still correct if we would keep $|k_T|$ finite in those. Thus we take the sum over helicities, and replace

$$\frac{|k_T|}{\kappa^+} A^+_n + \frac{|k_T|}{\kappa^-} A^-_n \to A^*_n \tag{199}$$

to find

$$\left|A^{*,+}_{n+1}\right|^2 + \left|A^{*,-}_{n+1}\right|^2 \to \frac{x^2}{|y|(x+y)^2} \frac{1}{|s_{Pr}|} \left\{2\left[1 - |y| \frac{s_{Pr}}{s_{kr}}\right] \left|A^*_n\right|^2 + y^2 \frac{s^2_{Pr}}{s^2_{kr}} \left[\left|A^+_n\right|^2 + \left|A^-_n\right|^2\right]\right\}. \tag{200}$$

Finally, the integral over the azimuthal angle of $k^\mu_T$ eliminates interference terms of the type

$$\frac{|k_T|^2}{(\kappa^\pm)^2} A^\pm_n \left(A^\mp_n\right)^\star \tag{201}$$

so eventually, we can write

$$\left|A^{*,+}_{n+1}\right|^2 + \left|A^{*,-}_{n+1}\right|^2 \to \frac{x^2}{|y|(x+y)^2} \frac{1}{|s_{Pr}|} \left\{2 - 2|y| \frac{s_{Pr}}{s_{kr}} + y^2 \frac{s^2_{Pr}}{s^2_{kr}}\right\} \left|A^*_n\right|^2\,. \tag{202}$$

# A Transverse momenta

For a transverse momentum $r_\perp^\mu$, we define

$$\kappa_r^\star = \frac{\langle P|\not{r}_\perp|\bar{P}]}{[P\bar{P}]} \quad , \quad \kappa_r = \frac{\langle \bar{P}|\not{r}_\perp|P]}{\langle \bar{P}P\rangle} \, . \tag{203}$$

Using polariztion vectors

$$e_+^\mu = \frac{\langle \bar{P}|\gamma^\mu|P]}{\sqrt{2}\langle \bar{P}P\rangle} \quad , \quad e_-^\mu = \frac{\langle P|\gamma^\mu|\bar{P}]}{\sqrt{2}[P\bar{P}]} \, , \tag{204}$$

satisfying $e_- \cdot e_- = e_+ \cdot e_+ = 0$, $e_- \cdot e_+ = -1$, we can then write

$$r_\perp^\mu = -\frac{\kappa_r^\star}{\sqrt{2}} \, e_+^\mu - \frac{\kappa_r}{\sqrt{2}} \, e_-^\mu \, . \tag{205}$$

The inner product of two transverse moment is given by

$$r_\perp \cdot q_\perp = \frac{e_- \cdot e_+}{2} (\kappa_r \kappa_q^\star + \kappa_r^\star \kappa_q) = -\frac{\kappa_r \kappa_q^\star + \kappa_r^\star \kappa_q}{2} \, . \tag{206}$$

# B Color

Using

$$T_{ij}^a T_{kl}^a = \frac{1}{2} \, \delta_{il} \delta_{jk} - \frac{1}{2N_c} \, \delta_{ij} \delta_{kl} \tag{207}$$

we get

$$\mathrm{Tr}(T^a X)\mathrm{Tr}(T^a Y) = \frac{1}{2}\mathrm{Tr}(XY) - \frac{1}{2N_c}\mathrm{Tr}(X)\mathrm{Tr}(Y) \tag{208}$$

$$\mathrm{Tr}(T^a X T^a Y) = \frac{1}{2}\mathrm{Tr}(X)\mathrm{Tr}(Y) - \frac{1}{2N_c}\mathrm{Tr}(XY) \tag{209}$$

$$\mathrm{Tr}(T^a X T^a) = \frac{N_c^2 - 1}{2N_c} \, \mathrm{Tr}(X) = C_F \, \mathrm{Tr}(X) \tag{210}$$

and then recursively

$$\begin{aligned} \mathrm{Tr}(T^a T^b X)\mathrm{Tr}(Y T^b T^a) &= \frac{1}{2}\mathrm{Tr}(T^b X Y T^b) - \frac{1}{2N_c}\mathrm{Tr}(T^b X)\mathrm{Tr}(Y T^b) \\ &= \frac{N_c^2 - 1}{4N_c} \, \mathrm{Tr}(XY) - \frac{1}{4N_c}\mathrm{Tr}(XY) + \frac{1}{4N_c^2}\mathrm{Tr}(X)\mathrm{Tr}(Y) \\ &= \frac{N_c^2 - 2}{4N_c} \, \mathrm{Tr}(XY) + \frac{1}{4N_c^2}\mathrm{Tr}(X)\mathrm{Tr}(Y) \end{aligned} \tag{211}$$

and

$$\begin{aligned}
\mathrm{Tr}(T^aT^bX)\mathrm{Tr}(T^aT^bY) &= \frac{1}{2}\mathrm{Tr}(T^bXT^bY) - \frac{1}{2N_c}\mathrm{Tr}(T^bX)\mathrm{Tr}(T^bY) \\
&= \frac{1}{4}\mathrm{Tr}(X)\mathrm{Tr}(Y) - \frac{1}{4N_c}\mathrm{Tr}(XY) - \frac{1}{4N_c}\mathrm{Tr}(XY) + \frac{1}{4N_c^2}\mathrm{Tr}(X)\mathrm{Tr}(Y) \\
&= -\frac{1}{2N_c}\,\mathrm{Tr}(XY) + \frac{N_c^2+1}{4N_c^2}\,\mathrm{Tr}(X)\mathrm{Tr}(Y)\ . \qquad (212)
\end{aligned}$$

# C Integrals

We want to calculate the integral

$$\begin{aligned}
\mathcal{F}(x_k, k_\perp; x_{\mathrm{thrs}}, \Lambda, \bar{x}_m) = &\int_0^1 \frac{dx_q}{2x_q}\int_0^{\bar{x}_m} d\bar{x}_q \int d^{2+\varepsilon} q_\perp \delta\left(\bar{x}_q - \frac{|q_\perp|^2}{s x_q \Lambda_k}\right) \\
&\times \int_{x_{\mathrm{thrs}}/\Lambda_k}^1 \frac{dx_r}{2x_r}\int_0^{\bar{x}_m} d\bar{x}_r \int d^{2+\varepsilon} r_\perp \delta\left(\bar{x}_r - \frac{|r_\perp|^2}{s x_r \Lambda_k}\right) \\
&\times \delta^2(k_\perp + q_\perp + r_\perp)\,\delta(x_q + x_r - 1)\,\mathcal{P}(x_q, q_\perp, x_r, r_\perp)\ , \qquad (213)
\end{aligned}$$

where $\mathcal{P}(x_q, q_\perp, x_r, r_\perp)$ has the general form

$$\mathcal{P}(x_q, q_\perp, x_r, r_\perp) = \mathcal{P}_{ab}(x_q, x_r)\,|q_\perp + r_\perp|^2 \left(\frac{C_{rq}}{|s_{r\bar{q}}||s_{q\bar{q}}|} + \frac{C_{r\bar{q}}\,x_q}{|s_{rq}||s_{q\bar{q}}|} + \frac{C_{q\bar{q}}\,x_r}{|s_{rq}||s_{r\bar{q}}|}\right)\ , \qquad (214)$$

where $\mathcal{P}_{ab}(x_q, x_r)$ is one of the collinear splitting functions and

$$s_{q\bar{q}} = -\frac{|q_\perp|^2}{x_q}\ \ ,\ \ s_{r\bar{q}} = -\frac{|r_\perp|^2}{x_r}\ \ ,\ \ s_{qr} = \frac{|q_\perp|^2}{x_q} + \frac{|r_\perp|^2}{x_r} - |q_\perp + r_\perp|^2\ . \qquad (215)$$

We find, up to powers of $1/\Lambda_k$,

$$\mathcal{F} = J(|k_\perp|^2, \varepsilon)\int_{x_{\mathrm{thrs}}/\Lambda_k}^1 dx_r\,\mathcal{P}_{ab}(1-x_r, x_r)\big(C_{rq} + C_{r\bar{q}}(1-x_r)^\varepsilon + C_{q\bar{q}}x_r^\varepsilon\big)\ , \qquad (216)$$

where

$$J(|k_\perp|^2, \varepsilon) = \frac{1}{4}\,\pi^{1+\varepsilon/2}\,\frac{\Gamma(1-\varepsilon/2)\Gamma^2(\varepsilon/2)}{\Gamma(\varepsilon)}\,|k_\perp|^\varepsilon = \pi^{1+\varepsilon/2}\,\frac{\Gamma(1-\varepsilon/2)\Gamma^2(1+\varepsilon/2)}{\Gamma(1+\varepsilon)}\,\frac{|k_\perp|^\varepsilon}{\varepsilon}\ . \qquad (217)$$

Realize that $\varepsilon = -2\varepsilon_{\mathrm{usual}}$.

## C.1 Quark-gluon

The coefficients are

$$C_{rq} = C_{r\bar{q}} = C_F + \frac{1}{2N_c} = \frac{N_c}{2}\quad ,\quad C_{q\bar{q}} = -\frac{1}{2N_c}\quad . \qquad (218)$$

The integrals are

$$\int_\xi^1 dx\, \frac{1+(1-x)^2+\delta\varepsilon x^2}{x} = 2\ln\frac{1}{\xi} - \frac{3}{2} + \frac{\delta}{2}\,\varepsilon + \mathcal{O}(\xi) \tag{219}$$

$$\begin{aligned}\int_\xi^1 dx\, \frac{1+(1-x)^2+\delta\varepsilon x^2}{x}(1-x)^\varepsilon &= 2\ln\frac{1}{\xi} - \frac{3}{2} + \mathcal{O}(\xi) \\ &\quad + \left(\frac{5+2\delta}{4} - \frac{\pi^2}{3} + \mathcal{O}(\xi)\right)\varepsilon + \mathcal{O}(\varepsilon^2)\end{aligned} \tag{220}$$

$$\begin{aligned}\int_\xi^1 dx\, \frac{1+(1-x)^2+\delta\varepsilon x^2}{x}\, x^\varepsilon &= 2\frac{1-\xi^\varepsilon}{\varepsilon} - 2\frac{1-\xi^{1+\varepsilon}}{1+\varepsilon} + (1+\delta\varepsilon)\frac{1-\xi^{2+\varepsilon}}{2+\varepsilon} \\ &= 2\frac{1-\xi^\varepsilon}{\varepsilon} - \frac{3}{2} + \mathcal{O}(\xi) + \left(\frac{7+2\delta}{4} + \mathcal{O}(\xi)\right)\varepsilon + \mathcal{O}(\varepsilon^2)\end{aligned} \tag{221}$$

where $\delta = 1/2$ (remember that $\varepsilon = -2\varepsilon_{\text{usual}}$). Ignoring $\mathcal{O}(\xi)$ and $\mathcal{O}(\varepsilon^2)$, we get

$$\mathcal{F} = \frac{N_c}{2}\,\pi^{1+\varepsilon/2}\, r_\Gamma\, |k_\perp|^\varepsilon \left[\frac{1}{\varepsilon}\left(4\ln\frac{\Lambda_k}{x_{\text{thrs}}} - 3\right) + \frac{5}{4} + \delta - \frac{\pi^2}{3} - \frac{1}{N_c^2}\left(2\frac{1-\xi^\varepsilon}{\varepsilon^2} - \frac{3}{2\varepsilon} + \frac{7}{4} + \frac{\delta}{2}\right)\right], \tag{222}$$

Where $\xi = x_{\text{thrs}}/\Lambda_k$. Replacing $\varepsilon \to -2\varepsilon$, we get

$$\mathcal{F} = \frac{N_c}{2}\,\pi^{1-\varepsilon}\, r_\Gamma\, |k_\perp|^{-2\varepsilon} \left[\frac{1}{\varepsilon}\left(\frac{3}{2} - 2\ln\frac{\Lambda_k}{x_{\text{thrs}}}\right) + \frac{5}{4} + \delta - \frac{\pi^2}{3} - \frac{1}{2N_c^2}\left(\frac{1-\xi^{-2\varepsilon}}{\varepsilon^2} + \frac{3}{2\varepsilon} + \frac{7}{2} + \delta\right)\right]. \tag{223}$$

## C.2 Gluon-gluon

Now we have to take into account that $x_q$ and $x_r$ should be equivalent, so the integral includes a $\theta(x_q - x_r)$, and there is a second contribution with $x_q$ and $x_r$ interchanged (so $x_q$ gets the lower limit $x_{\text{thrs}}/\Lambda_k$). The coefficients are

$$C_{rq} = C_{r\bar{q}} = C_{q\bar{q}} = \frac{N_c}{2}\,. \tag{224}$$

Using

$$\frac{1+x^4+(1-x)^4}{x(1-x)} = 2\left(\frac{1}{x} + \frac{1}{1-x} - 2 + x - x^2\right) \tag{225}$$

we find the integrals

$$\int_\xi^{1/2} dx\, \frac{1+x^4+(1-x)^4}{x(1-x)} = 2\ln\frac{1}{\xi} - \frac{11}{6} + \mathcal{O}(\xi) \tag{226}$$

$$\int_\xi^{1/2} dx\, \frac{1+x^4+(1-x)^4}{x(1-x)}(1-x)^\varepsilon = 2\ln\frac{1}{\xi} - \frac{11}{6} + \mathcal{O}(\xi) + \left(\frac{131}{72} - \frac{11\ln 2}{6} - \frac{\pi^2}{6} + \mathcal{O}(\xi)\right)\varepsilon + \mathcal{O}(\varepsilon^2) \tag{227}$$

$$\int_\xi^{1/2} dx\, \frac{1+x^4+(1-x)^4}{x(1-x)}\, x^\varepsilon = 2\frac{1-\xi^\varepsilon}{\varepsilon} - \frac{11}{6} + \mathcal{O}(\xi) + \left(\frac{137}{72} + \frac{11\ln 2}{6} - \frac{\pi^2}{6} + \mathcal{O}(\xi)\right)\varepsilon + \mathcal{O}(\varepsilon^2) \tag{228}$$

Now we find

$$\mathcal{F} = \frac{N_c}{2}\,\pi^{1+\varepsilon/2}\, r_\Gamma\, |k_\perp|^\varepsilon \left[2\frac{1-\xi^\varepsilon}{\varepsilon^2} + \frac{1}{\varepsilon}\left(4\ln\frac{\Lambda_k}{x_{\rm thrs}} - \frac{11}{2}\right) + \frac{67}{18} - \frac{\pi^2}{3}\right], \tag{229}$$

where $\xi = x_{\rm thrs}/\Lambda_k$. Putting $\varepsilon \to -2\varepsilon$ again, we get

$$\mathcal{F} = \frac{N_c}{2}\,\pi^{1-\varepsilon}\, r_\Gamma\, |k_\perp|^{-2\varepsilon} \left[\frac{1-\xi^{-2\varepsilon}}{2\varepsilon^2} + \frac{1}{\varepsilon}\left(-2\ln\frac{\Lambda_k}{x_{\rm thrs}} + \frac{11}{4}\right) + \frac{67}{18} - \frac{\pi^2}{3}\right]. \tag{230}$$

Including the second half of phase space, we find

$$\mathcal{F} = \frac{N_c}{2}\,\pi^{1-\varepsilon}\, r_\Gamma\, |k_\perp|^{-2\varepsilon} \left[\frac{1-\xi^{-2\varepsilon}}{\varepsilon^2} + \frac{1}{\varepsilon}\left(-4\ln\frac{\Lambda_k}{x_{\rm thrs}} + \frac{11}{2}\right) + \frac{67}{9} - \frac{2\pi^2}{3}\right]. \tag{231}$$

Compare to the quark-gluon case with $\delta = 1/2$:

$$\mathcal{F} = \frac{N_c}{2}\,\pi^{1-\varepsilon}\, r_\Gamma\, |k_\perp|^{-2\varepsilon} \left[\frac{1}{\varepsilon}\left(\frac{3}{2} - 2\ln\frac{\Lambda_k}{x_{\rm thrs}}\right) + \frac{7}{4} - \frac{\pi^2}{3} - \frac{1}{2N_c^2}\left(\frac{1-\xi^{-2\varepsilon}}{\varepsilon^2} + \frac{3}{2\varepsilon} + 4\right)\right]. \tag{232}$$

# D Overall factors

From Ellis-Sexton, we find and extra factors of 2 for the high-energy splitting functions, so

$$\mathcal{P}_{\rm aux-q}(x_q, q_\perp, x_r, r_\perp) = N_c\left[|s_{qr}| + x_q|s_{r\bar{q}}| - \frac{x_r|s_{q\bar{q}}|}{N_c^2}\right]\frac{1+x_q^2}{x_r}\,\frac{|q_\perp + r_\perp|^2}{|s_{q\bar{q}}||s_{r\bar{q}}||s_{qr}|} \tag{233}$$

$$\mathcal{P}_{\rm aux-g}(x_q, q_\perp, x_r, r_\perp) = N_c\,\frac{1+x_q^4+x_r^4}{x_q x_r}\left(|s_{rq}| + x_q|s_{r\bar{q}}| + x_r|s_{q\bar{q}}|\right)\frac{|q_\perp + r_\perp|^2}{|s_{qr}||s_{q\bar{q}}||s_{r\bar{q}}|}. \tag{234}$$

Furthermore, they each get factors from the coupling and the phase space (our differential integration volume was just $d^{3-2\varepsilon}r$)

$$g^2\,\mu^{2\varepsilon}\,\frac{1}{(2\pi)^{3-2\varepsilon}} = 4\pi\alpha_s\,\mu^{2\varepsilon}\,\frac{(2\pi)^{2\varepsilon}}{(2\pi)^3} = \frac{\alpha_s}{2\pi}\,\mu^{2\varepsilon}\,\frac{(4\pi)^\varepsilon}{\pi^{1-\varepsilon}} \tag{235}$$

For the integrated function, we then find

$$\frac{\mathcal{F}_{\text{aux-q}}}{\mathcal{K}} = \frac{1}{\varepsilon}\left(-2\ln\frac{\Lambda_k}{x_{\text{thrs}}} + \frac{3}{2}\right) - \frac{\pi^2}{3} + \frac{5}{4} + \delta - \frac{1}{2N_c^2}\left(\frac{1-\xi^{-2\varepsilon}}{\varepsilon^2} + \frac{3}{2\varepsilon} + \frac{7}{2} + \delta\right) \tag{236}$$

$$\frac{\mathcal{F}_{\text{aux-g}}}{\mathcal{K}} = \frac{1}{\varepsilon}\left(-4\ln\frac{\Lambda_k}{x_{\text{thrs}}} + \frac{11}{2}\right) - \frac{2\pi^2}{3} + \frac{67}{9} + \frac{1-\xi^{-2\varepsilon}}{\varepsilon^2}\,, \tag{237}$$

with

$$\mathcal{K} = \frac{\alpha_s}{2\pi}(4\pi)^{\varepsilon}\, r_\Gamma\, N_c \left(\frac{\mu^2}{|k_\perp|^2}\right)^{\varepsilon}\,. \tag{238}$$

Remember that $\delta = 1/2$ makes the splitting function $(4-2\varepsilon)$-dimensional. For the virtual contributions proportional to $(\mu^2/|k_\perp|^2)^\varepsilon$, we find (with $n_f = 0$)

$$\frac{\mathcal{V}_{\text{aux-q}}}{\mathcal{K}} = \frac{1}{\varepsilon}\left(2\ln\Lambda + \frac{13}{6} + \Xi\right) + \frac{\pi^2}{3} + \frac{83}{18} - \frac{\delta_R}{6} + \frac{1}{N_c^2}\left(\frac{1}{\varepsilon^2} + \frac{3}{2\varepsilon} + \frac{7}{2} + \frac{\delta_R}{2}\right) \tag{239}$$

$$\frac{\mathcal{V}_{\text{aux-g}}}{\mathcal{K}} = \frac{1}{\varepsilon}\left(2\ln\Lambda + \Xi\right) + \frac{\pi^2}{3} - \frac{1}{\varepsilon^2} \tag{240}$$

where $\lim_{|k_T|\to 0}\Xi = -i\pi$, and $\delta_R = 1$ for the 't Hooft-Veltman scheme. If we include an extra factor of 2 for $\mathcal{F}_{\text{aux-q}}$, then

$$\left(\frac{\mathcal{V}_{\text{aux-g}}}{\mathcal{K}} + \frac{\mathcal{F}_{\text{aux-g}}}{\mathcal{K}}\right) - \left(\frac{\mathcal{V}_{\text{aux-q}}}{\mathcal{K}} + \frac{2\mathcal{F}_{\text{aux-q}}}{\mathcal{K}}\right) = \frac{1}{3\varepsilon} - \frac{1}{2}\,. \tag{241}$$

# E Collinear splitting functions

The collinear limit of a momentum pair $p_i^\mu, p_j^\mu$ in an amplitude is usually derived by factoring out the obvious singular part, and then replacing

$$p_i^\mu \to (1-z)p_{ij}^\mu \quad , \quad p_j^\mu \to z p_{ij}^\mu\,. \tag{242}$$

Consider for example the MHV amplitude

$$A_n\big(1^-,2^+,3^-,4^+,\ldots,n^+\big) = i\frac{\langle p_1p_3\rangle^4}{\langle p_1p_2\rangle\langle p_2p_3\rangle\langle p_3p_4\rangle\cdots\langle p_{n-1}p_n\rangle\langle p_np_1\rangle} \tag{243}$$

which is singular when $p_1^\mu, p_2^\mu$ become collinear. Performing the procedure mentioned above, we get

$$\begin{aligned} A_n\big(1^-,2^+,3^-,4^+,\ldots,n^+\big) &\to \frac{1}{\langle p_1p_2\rangle}\, i\frac{(1-z)^2\langle p_{12}p_3\rangle^4}{\sqrt{z}\langle p_{12}p_3\rangle\langle p_3p_4\rangle\cdots\langle p_{n-1}p_n\rangle\sqrt{1-z}\langle p_np_{12}\rangle} \\ &= \frac{1}{\langle p_1p_2\rangle}\,\frac{(1-z)^2}{\sqrt{z(1-z)}}\,A_{n-1}\big((12)^-,3^-,4^+,\ldots,n^+\big) \end{aligned} \tag{244}$$

which corresponds to Eq.(6.15) in [4] (because $A_{n-1}\big((12)^+,3^-,4^+,\ldots,n^+\big) = 0$).

Let us consider a slightly more general procedure, in which we take the limits $p_i^\mu \to x p_{ij}^\mu$, $p_j^\mu \to y p_{ij}^\mu$, and then correct the factorized amplitude so it contains momentum $(x+y)p_{ij}^\mu$. For the example above we get

$$\begin{aligned} A_n\big(1^-,2^+,3^-,4^+,\dots,n^+\big) &\to \frac{1}{\langle p_1 p_2\rangle}\,\frac{x^2}{\sqrt{xy}}\,i\frac{\langle p_{12}p_3\rangle^4}{\langle p_{12}p_3\rangle\langle p_3p_4\rangle\cdots\langle p_{n-1}p_n\rangle\langle p_n p_{12}\rangle} \\ &= \frac{1}{\langle p_1 p_2\rangle}\,\frac{x^2}{\sqrt{xy}(x+y)}\,i\frac{\langle (x+y)p_{12}p_3\rangle^4}{\langle (x+y)p_{12}p_3\rangle\langle p_3p_4\rangle\cdots\langle p_{n-1}p_n\rangle\langle p_n (x+y)p_{12}\rangle} \\ &= \frac{1}{\langle p_1 p_2\rangle}\,\frac{x^2}{\sqrt{xy}(x+y)}\,A_{n-1}\big((x+y)p_{12}^-,3^-,4^+,\dots,n^+\big)\ . \end{aligned} \tag{245}$$

We get the previous formula by simply setting $x = 1-z$ and $y = z$.

When 1 refers to an initial state and 2 to a final state, the value of $y$ becomes negative, and the question occurs what happens with the square root. This is determined by the convention for the spinors, which for us says that

$$\big|-|y|p\big\rangle = \sqrt{|y|}\,\big|p\big\rangle \quad , \quad \big|-|y|p\big] = -\sqrt{|y|}\,\big|p\big]\ . \tag{246}$$

For squared amplitudes summed over helicities, the gluon splitting function becomes

$$\mathcal{Q}_{gg}(x,y) = \frac{x^4+y^4+(x+y)^4}{|xy|(x+y)^2} = \frac{\big|\mathcal{P}_{gg}(1+y/x)\big|}{1+y/x}\ . \tag{247}$$

Here we see that negative $y$ would make $\mathcal{Q}_{gg}$ negative without explicitly taking absolute value. This could make sense if we would choose the initial-state momentum involved in the collinear limit to have negative energy so that also its scalar product with the final-state momentum involved would be negative, but we prefer not to and force these quantities to be positive at the cross-section level. Finally, after some algebra one indeed finds,

$$\mathcal{Q}_{gg}(1-z,z) = \mathcal{P}_{gg}(z)\ . \tag{248}$$

In the following, we denote

$$(ab)_- = \langle ab\rangle \quad \text{and} \quad (ab)_+ = [ab] \quad . \tag{249}$$

For the splitting of a quark or anti-quark into a quark or anti-quark and a gluon we have

$$A_n(q^\pm,g^\pm,\dots) \xrightarrow{p_q\to zp_q\,,\,p_g\to(1-z)p_q} \frac{\pm 1}{\sqrt{1-z}\,(p_q p_g)_\mp}\,A_{n-1}(q^\mp,\dots) \tag{250}$$

$$A_n(q^\pm,g^\mp,\dots) \xrightarrow{p_q\to zp_q\,,\,p_g\to(1-z)p_q} \frac{\mp z}{\sqrt{1-z}\,(p_q p_g)_\pm}\,A_{n-1}(q^\mp,\dots) \tag{251}$$

$$A_n(\dots,g^\mp,\bar q^\mp) \xrightarrow{p_{\bar q}\to zp_{\bar q}\,,\,p_g\to(1-z)p_{\bar q}} \frac{\mp 1}{\sqrt{1-z}\,(p_g p_{\bar q})_\pm}\,A_{n-1}(\dots,\bar q^\pm) \tag{252}$$

$$A_n(\dots,g^\pm,\bar q^\mp) \xrightarrow{p_{\bar q}\to zp_{\bar q}\,,\,p_g\to(1-z)p_{\bar q}} \frac{\pm z}{\sqrt{1-z}\,(p_g p_{\bar q})_\mp}\,A_{n-1}(\dots,\bar q^\pm) \tag{253}$$

These lead to a splitting function for helicity-summed squared amplitudes given by

$$\mathcal{P}_{qq}(z) = \frac{1+z^2}{1-z} \,. \tag{254}$$

The more general amplitude-level splittings for $p_{q/\bar{q}} \to x p_{q/\bar{q}}$ and $p_g \to y p_{q/\bar{q}}$ are given by

$$\frac{\pm 1}{\sqrt{y}\,(p_q p_g)_\mp} \,, \quad \frac{\mp x}{\sqrt{y}\,(x+y)(p_q p_g)_\pm} \,, \quad \frac{\mp 1}{\sqrt{y}\,(p_g p_{\bar{q}})_\pm} \,, \quad \frac{\pm x}{\sqrt{y}\,(x+y)(p_g p_{\bar{q}})_\mp} \,. \tag{255}$$

leading to the splitting function

$$\mathcal{Q}_{qq}(x,y) = \frac{1}{|y|}\left(1 + \frac{x^2}{(x+y)^2}\right) = \frac{\left|\mathcal{P}_{qq}(1+y/x)\right|}{x(1+y/x)^2} \,. \tag{256}$$

# F Collinear factorization

Consider an $(n+1)$-particle phase space with one momentum singled out and denoted $r^\mu$, and decomposed as

$$r^\mu = x_r P^\mu + \bar{x}_r \bar{P}^\mu + r_\perp^\mu \,. \tag{257}$$

The other $n$ momenta are collectively denoted $\Phi_n$, and their sum is $Q_n^\mu$. We have

$$\begin{aligned} d\sigma_n = \int_0^1 dx\, f(x) \int_0^1 d\bar{x}\, f(\bar{x}) \int_0^x dx_r \int_0^{\bar{x}} d\bar{x}_r \int d^{2-2\varepsilon} r_\perp \, \frac{s}{2} \delta\big(s x_r \bar{x}_r - |r_\perp|^2\big) \\ \times \, [d\Phi_n]\, \delta^{4-2\varepsilon}\big(xP + \bar{x}\bar{P} - Q_n - r\big)\, M_{n+1}(-xP, r) \,. \end{aligned} \tag{258}$$

We suppress the dependence of $M_{n+1}$ on $\bar{x}\bar{P}$ and $\Phi_n$ in the notation. For the matrix element, we want all momenta to be outgoing, explaining the minus sign in $-xP$. We are interested in the situation in which $|r_\perp|$ goes to zero while $x_r$ does not, implying that also $\bar{x}_r$ goes to zero. The matrix element will typically behave as

$$M_{n+1}\big(-xP, r\big) \overset{|r_\perp| \to 0}{\longrightarrow} \frac{C}{2xP\cdot r}\, \mathcal{Q}(x, -x_r)\, M_n\big(-(x-x_r)P\big) \,, \tag{259}$$

where $C$ contains the color factor, the coupling, etc. The cross section at the limit for the matrix element and phase space becomes

$$\begin{aligned} d\sigma_n^{\text{lim}} = \int_0^1 dx\, f(x) \int_0^1 d\bar{x}\, f(\bar{x}) \int_0^x dx_r \int_0^{\bar{x}} d\bar{x}_r \int d^{2-2\varepsilon} r_\perp \, \frac{s}{2} \delta\left(s x_r \bar{x}_r - |r_\perp|^2\right) \\ \times \frac{C}{2xP\cdot r}\, \mathcal{Q}(x, -x_r)\, [d\Phi_n]\, \delta^{4-2\varepsilon}\big((x-x_r)P + \bar{x}\bar{P} - Q_n\big)\, M_n\big(-(x-x_r)P\big) \,. \end{aligned} \tag{260}$$

Write $2P\cdot r = s\bar{x}_r$

$$\begin{aligned} d\sigma_n^{\text{lim}} = \int_0^1 dx\, f(x) \int_0^1 d\bar{x}\, f(\bar{x}) \int_0^x dx_r \int_0^{\bar{x}} d\bar{x}_r \int d^{2-2\varepsilon} r_\perp \, \frac{s}{2} \delta\left(s x_r \bar{x}_r - |r_\perp|^2\right) \frac{C}{s\bar{x}_r} \frac{\mathcal{Q}(x, -x_r)}{x} \\ \times \, [d\Phi_n]\, \delta^{4-2\varepsilon}\big((x-x_r)P + \bar{x}\bar{P} - Q_n\big)\, M_n\big(-(x-x_r)P\big) \,. \end{aligned} \tag{261}$$

Execute the $\bar{x}_r$ integration

$$d\sigma_n^{\rm lim} = \int_0^1 dx\, f(x) \int_0^1 d\bar{x}\, f(\bar{x}) \int_0^x dx_r \int d^{2-2\varepsilon} r_\perp \frac{1}{2}\theta\left(\bar{x} - \frac{|r_\perp|^2}{x_r s}\right) \frac{C}{|r_\perp|^2} \frac{\mathcal{Q}(x,-x_r)}{x} \times [d\Phi_n]\, \delta^{4-2\varepsilon}\big((x-x_r)P + \bar{x}\bar{P} - Q_n\big)\, M_n\big(-(x-x_r)P\big)\,. \quad (262)$$

Substitute $x_r \to (1-z)x$:

$$d\sigma_n^{\rm lim} = \int_0^1 dx\, f(x) \int_0^1 d\bar{x}\, f(\bar{x}) \int_0^1 dz \int d^{2-2\varepsilon} r_\perp \frac{1}{2}\theta\left(\bar{x} - \frac{|r_\perp|^2}{(1-z)xs}\right) \frac{C}{|r_\perp|^2} \frac{\mathcal{Q}\big(x,-x(1-z)\big)}{x} \times [d\Phi_n]\, \delta^{4-2\varepsilon}\big(zxP + \bar{x}\bar{P} - Q_n\big)\, M_n(-zxP)\,. \quad (263)$$

Performing the $r_\perp$ integral and using Eq. (247), we get

$$d\sigma_n^{\rm lim} = \int_0^1 dx\, f(x) \int_0^1 d\bar{x}\, f(\bar{x}) \int_0^1 dz\, \frac{1}{2} \frac{2\pi^{1-\varepsilon}}{\Gamma(1-\varepsilon)} \frac{\big(sx\bar{x}(1-z)\big)^{-\varepsilon}}{-2\varepsilon}\, C\, \frac{\mathcal{P}_{gg}(z)}{zx} \times [d\Phi_n]\, \delta^{4-2\varepsilon}\big(zxP + \bar{x}\bar{P} - Q_n\big)\, M_n(-zxP)\,. \quad (264)$$

# G Feynman gauge

In the Feynman gauge we have $K_{i,j}\cdot\mathcal{J}_{i,j} = 0$ and $K_{i,j}\cdot\tilde{\mathcal{J}}_{i,j} = 0$ for any off-shell current, and the general recursion reduces to

$$\tilde{\mathcal{J}}^\mu_{1,j} = \frac{i}{\sqrt{2}} \sum_{k=1}^{j-1} \Big[(K_{1,k} - K_{k+1,j})^\mu (\mathcal{J}_{1,k}\cdot\mathcal{J}_{k+1,j}) + 2\big(K_{k+1,j}\cdot\mathcal{J}_{1,k}\big)\mathcal{J}^\mu_{k+1,j} - 2\big(K_{1,k}\cdot\mathcal{J}_{k+1,j}\big)\mathcal{J}^\mu_{1,k}\Big] + \big(\text{four-point vertex terms}\big)\,, \quad (265)$$

and for gluon 1 having divergent momentum Eq. (131), it reduces further to

$$\tilde{\mathcal{J}}^\mu_{1,j} = \frac{i}{\sqrt{2}} \sum_{k=1}^{n-1} \Big[\Lambda P^\mu (\mathcal{J}_{1,k}\cdot\mathcal{J}_{k+1,j}) - 2\Lambda\big(P\cdot\mathcal{J}_{k+1,j}\big)\mathcal{J}^\mu_{1,k}\Big] \;+\; \mathcal{O}\big(\Lambda^0\big)\,, \quad (266)$$

which can easily be checked to be consistent with

$$\mathcal{J}^\mu_{1,j} = A_{1,j}\,\mathcal{J}^\mu_1 + B_{1,j}P^\mu + \mathcal{O}\big(\Lambda^{-1}\big)\,, \quad (267)$$

for some coefficients $A_{1,j}$ and $B_{1,j}$ that are $\mathcal{O}\big(\Lambda^0\big)$. Of course $A_{1,1} = 1$ and $B_{1,1} = 0$, and

$$\mathcal{J}^\mu_1 = \tilde{\mathcal{J}}^\mu_1 = \varepsilon^\mu(K_1)\,. \quad (268)$$

Only the second term in Eq. (**??**) with $\mathcal{J}^\mu_{1,k}$ contributes to the coefficients $A_{1,j}$. We can project onto the $\mathcal{J}^\mu_1$ component of $\mathcal{J}^\mu_{1,j}$ by contracting with a vector $\ell^\mu$ such that

$$\ell\cdot P = 0 \quad \text{while} \quad \ell\cdot\mathcal{J}_1 \neq 0 \quad (269)$$

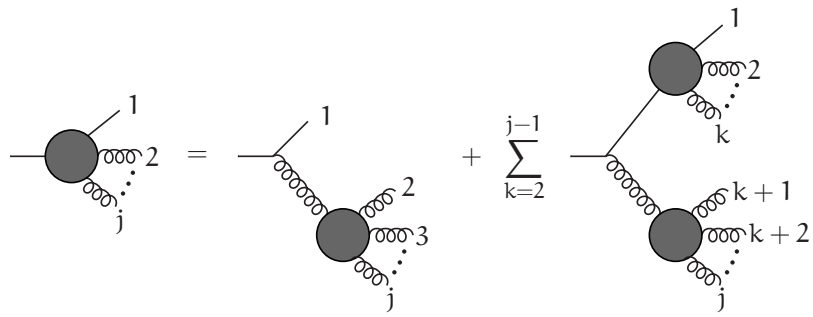



Figure 2: Berends-Giele recursive relation for quark-type line.

to find

$$\ell\cdot\tilde{\mathcal{J}}_{1,j} = -\sqrt{2}\,i\Lambda\sum_{k=1}^{j-1}\big(P\cdot\mathcal{J}_{k+1,j}\big)(\ell\cdot\mathcal{J}_1)A_{1,k} \;+\; \mathcal{O}\big(\Lambda^0\big)\;, \tag{270}$$

that is

$$\begin{aligned}(\ell\cdot\mathcal{J}_1)A_{1,j} = \ell\cdot\mathcal{J}_{1,j} &= -i\frac{\ell\cdot\tilde{\mathcal{J}}_{1,j}}{2\Lambda P\cdot K_{1,j}} \;+\; \mathcal{O}\big(\Lambda^{-1}\big)\\ &= \frac{-\sqrt{2}}{2P\cdot K_{1,j}}\sum_{k=1}^{j-1}\big(P\cdot\mathcal{J}_{k+1,j}\big)(\ell\cdot\mathcal{J}_1)A_{1,k} \;+\; \mathcal{O}\big(\Lambda^{-1}\big)\;.\end{aligned} \tag{271}$$

Then, defining

$$\tilde{A}_{1,j} = -2i(P\cdot K_{1,j})A_{1,j} \quad \text{for } j>1 \quad , \quad \tilde{A}_{1,1} = A_{1,1} = 1\;, \tag{272}$$

we find

$$\tilde{A}_{1,j} = i\sqrt{2}(P\cdot\mathcal{J}_{2,j})\tilde{A}_{1,1} - \sqrt{2}\sum_{k=2}^{j-1}\frac{P\cdot\mathcal{J}_{k+1,j}}{2P\cdot K_{1,k}}\,\tilde{A}_{1,k} \;+\; \mathcal{O}\big(\Lambda^{-1}\big)\;. \tag{273}$$

So we see that the coefficients $\tilde{A}_{1,j}$ follow the recursion of an eikonal quark line as in Fig. 2, with quark-gluon vertex $i\sqrt{2}P^\mu$ and quark-line propagator $i/(2P\cdot K)$. The term for $k=1$ is written separately because it must not include the eikonal propagator. The coefficients $\tilde{A}_{1,j}$ were of course defined to be the natural coefficients for $\tilde{\mathcal{J}}_{1,j}$, and we can similarly define $\tilde{B}_{1,j}$, so that

$$\tilde{\mathcal{J}}^\mu_{1,j} = -\Lambda\tilde{A}_{1,j}\mathcal{J}^\mu_1 - \Lambda\tilde{B}_{1,j}P^\mu + \mathcal{O}\big(\Lambda^0\big)\;. \tag{274}$$

The minus sign comes from the difference in propagator definitions, $-i/K^2$ for gluons, while $+i/(2P\cdot K)$ for eikonal lines. We conclude that if $n+1$ gluons satisfy momentum conservation $\sum_{i=1}^{n+1}K_i = 0$ and number 1 and number $n+1$ have a divergent coefficient for $P^\mu$, then

$$\begin{aligned}&\tilde{A}_{1,n} \text{ gives the amplitude for } n \text{ gluons, with num-}\\ &\text{ber 1 being space-like with momentum } K^\mu_1 + K^\mu_{n+1},\end{aligned} \tag{275}$$

which is the coefficient of $\mathcal{J}_1^\mu$ in $\tilde{\mathcal{J}}_{1,n}^\mu$ divided by $\Lambda$. It can be extracted from $\tilde{\mathcal{J}}_{1,n}^\mu$ by contracting with $\mathcal{J}_{n+1}^\mu$. Then $\mathcal{J}_{n+1}\cdot P = \mathcal{O}\big(\Lambda^{-1}\big)$, and $\mathcal{J}_{n+1}\cdot\mathcal{J}_1 = -1 + \mathcal{O}\big(\Lambda^{-1}\big)$ if it has the opposite helicity from $\mathcal{J}_1^\mu$. The connection with the result in the axial gauge can be established by imagining that in Eq. (**??**) $P\cdot\mathcal{J}_{2,j}^\mu \neq 0$ only for $j = n$, so that $\mathcal{J}_{n+1}\cdot\tilde{\mathcal{J}}_{1,n} = \Lambda\tilde{A}_{1,n} = \Lambda i\sqrt{2}(P\cdot\mathcal{J}_{2,n}) + \mathcal{O}\big(\Lambda^0\big)$, just like in Eq. (144).

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